

The Portfolio-Driven Disposition Effect*

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Abstract: The disposition effect for a stock significantly weakens if the portfolio is at a gain, but is large when it is at a loss. We find this portfolio-driven disposition effect (PDDE) in four independent settings: US and Chinese archival data, as well as US and Chinese experiments. The PDDE is robust to a variety of controls in regression specifications and is not explained by extreme returns, portfolio rebalancing, tax considerations, or investor heterogeneity. Our evidence suggests investors form mental frames at the stock and portfolio level and these frames combine to generate the PDDE.

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I. Introduction

There is perhaps no more robust trading phenomenon than the disposition effect, the observation that investors are more likely to sell an asset when it is at a gain than when it is at a loss (Shefrin and Statman, 1985). The disposition effect has been documented among US retail investors (Odean, 1998), foreign retail investors (Grinblatt and Keloharju, 2001; Frydman and Wang, 2020), institutional investors (Shapira and Venezia, 2001), homeowners (Genesove and Mayer, 2001), corporate executives (Heath, Huddart, and Lang, 1999), and in experimental settings (Frydman, Hartzmark, and Solomon, 2018; Weber and Camerer, 1998).

Standard explanations for the disposition effect – such as tax considerations, portfolio rebalancing, and informed trading – have been proposed and dismissed (Odean, 1998), leaving explanations that rely on investor preferences.¹ Models which attempt to explain the disposition effect often have investors with preferences over some subset of their wealth such as an individual stock, which Thaler et al. (1997) call “narrow framing.” For example, Barberis and Xiong (2009, 2012) and Ingersoll and Jin (2013) show that if an investor has preferences defined over realized stock-level gains and losses, she will predictably exhibit a disposition effect.

While much of the empirical and theoretical work related to the disposition effect focuses on individual assets, most households hold a portfolio of assets. This paper then asks a simple question: does the disposition effect operate independently for each individual asset, or does it depend on the portfolio as a whole? In doing so, we ask the related question of whether investors have preferences over individual stocks, the portfolio as a whole, or both.

To illustrate the idea, consider an investor with three stocks: X_1 , X_2 , and X_3 . The disposition effect says $\Pr(X_i \text{ is sold} \mid X_i \text{ is at a gain}) > \Pr(X_i \text{ is sold} \mid X_i \text{ is at a loss})$ for all i . If the investor has preferences over each individual stock, then we would expect those three

¹ Belief-based interpretations have also been proposed. Odean (1998) discusses that the disposition effect is consistent with investors having an irrational belief in price mean reversion. Ben-David and Hirshleifer (2012) argue that belief-based interpretations can offer a possible explanation for the V-shapes of both the selling and buying schedules that they document.

probabilistic statements to be independent of each other. However, if preferences depend in part on portfolio performance, then we would expect the disposition effect for Stock X_1 to depend on the state of the remaining portfolio (X_2 and X_3).

The latter is precisely what we find in the data. Whether we examine 78,000 households in the Barber and Odean (2000) dataset, 97,000 investors in a Chinese brokerage dataset, 2,300 US participants in an Amazon Mechanical Turk (“MTurk”) trading-game experiment, or 800 experimental participants at a Chinese university, all the data tell the same story: an investor’s disposition effect is large when her portfolio is at a loss and significantly smaller when it is at a gain.

To illustrate the main finding, consider the probability of selling a given stock among the four possible (Stock, Portfolio) conditions: (Gain, Gain), (Gain, Loss), (Loss, Gain) and (Loss, Loss). When we calculate simple univariate statistics for each of these in the Barber and Odean dataset, we find that the probability of selling in the (Gain, Gain) and (Loss, Gain) conditions are nearly equal: in other words, there is almost no disposition effect when the portfolio is at a gain. Given how pervasive the disposition effect is, it is surprising to find that the disposition effect largely disappears among the 61% of observations in which portfolios are up in the Barber and Odean (2000) dataset.

Because it is well known that there is a disposition effect in the Barber and Odean dataset, there must be a large effect among the remaining 39% of the data when portfolios are at a loss. This is precisely what we find: the probability of selling in the (Gain, Loss) condition is nearly twice as large as that of the (Loss, Loss) condition. We call this the *portfolio-driven disposition effect* (PDDE).

We document this relationship between the performance of an investor’s portfolio and her tendency to exhibit a disposition effect in both univariate analysis and hazard regressions with a host of controls. Perhaps the cleanest way to see our findings is via a matched-sample analysis. More specifically, we compare selling decisions across investors made on the same day for the

same stock that was also purchased on the same day. In other words, our identification comes exclusively from the fact that different investors face different portfolio-level capital gains due to the other stocks in their portfolios. The results are very similar to those from the baseline analysis.

The PDDE is not a repackaging of earlier research on the disposition effect. Specifically, we show that it is distinct from the rank effect documented by Hartzmark (2015), and it is not explained by tax considerations, portfolio rebalancing, or investor heterogeneity in the disposition effect.

The evidence is most consistent with investors having at least two frames – one at the stock level and one at the portfolio level - when making their trading decisions. While prior research on the disposition effect has established narrow framing at the stock level, here we provide empirical evidence of an additional frame at the portfolio level which interacts with the stock-level frame, resulting in the PDDE. To do this, we exploit the fact that a focal asset's membership in a portfolio will be a function of how similar the other assets in the portfolio are. Similarity has long been thought to be a defining characteristic of how an individual creates her mental account (Goldstone, 1994; Kruschke, 1992; Nosofsky, 1986). As Evers, Imas, and Kang (2022) put it: “when outcomes are perceived to be similar, they are categorized together, assigned to the same mental account and evaluated jointly.” Thus, when considering focal Stock X, if investors frame at the portfolio level, then we should expect a stronger PDDE when defining a portfolio with assets most similar to Stock X. For example, consider an investor that owns 3 assets: Stock X, Stock Y, and a house. When considering focal Stock X, similarity would dictate that Stock Y is more likely to be placed in the same mental account as Stock X, and thus will be more influential to the trading decisions of Stock X than the house.

With this in mind, we perform two tests based on similarity. First, we exploit the fact that a single household can have multiple accounts from the same discount brokerage. Similarity would predict that two stocks in the same account are more likely to be considered in the same portfolio while two stocks from different accounts are less likely to be, even though all stocks in

the brokerage accounts contribute to the household's wealth. We find evidence of PDDE moderation following dissimilarity: stocks held by the same account as the focal stock generate a PDDE which is 21% larger than that of stocks held in the same household but a different account than the focal stock. This pattern is also robust when restricting to one-adult households, where the size of the effect is 28%.

Second, rather than measuring the similarity of stocks across brokerage accounts within the same household, we exploit the characteristics of the individual assets within a single brokerage account. Specifically, we sort investors' assets into US common stocks, foreign stocks, open-end mutual funds, options, and other stock-type securities (such as closed-end funds and preferred stock). When the focal stock is a US common stock, the PDDE shrinks as the source of the portfolio capital gain bears less resemblance to US common stocks. For example, the moderating effect of one unit of capital gains generated by other US common stocks in the portfolio is 2.7 times as large as that of foreign stocks, 3.2 times as large as that of other stock-type securities, and 3.6 times as large as that of mutual funds. In other words, as a stock in the portfolio looks less similar to the focal stock, its contribution to the PDDE wanes.

The PDDE has important downstream consequences for aggregate behavior, prices, and investor welfare. When aggregate market indices rise (fall), a greater fraction of investors will have portfolios at a gain (loss), and so the PDDE predicts aggregate countercyclicality in the disposition effect. We confirm this in both the US and Chinese brokerage data: following a bull market, the disposition effect falls, but following a bear market, it rises. Our evidence supports the view that investors engage in more heuristic-like behavior in bad times (Mullainathan and Shafir, 2013; Schilbach, Schofield, and Mullainathan, 2016).

Moreover, our evidence that investors frame at both the individual stock and portfolio level has the potential to explain stock return patterns that existing single-frame models of investor behavior have difficulty predicting. For example, Barberis and Huang (2001) develop two models of framing, one in which investors frame at the individual stock level and another where

they frame at the total portfolio level. Each of their models can explain some well-known empirical patterns, and they conclude that a superior model of investor behavior would include both stock-level framing and broader forms of framing.² Our evidence here via the PDDE is that investors indeed frame at both the stock and portfolio level.

Our paper is organized as follows. We describe our data and methodology in Section II. In Section III, we introduce the PDDE and show that it is a robust phenomenon. In Section IV, we show that the PDDE is not explained by prior research. Section V provides direct evidence of a multiple-frame explanation for the PDDE. Section VI concludes.

II. Data and Methodology

A. Retail Brokerage Data

We begin with the large US discount broker dataset utilized by Barber and Odean (2000). The raw data include trading activity for roughly 78,000 households with roughly 158,000 accounts between January 1991 and November 1996. Following Odean (1998), we restrict our main analyses to the US common stocks because the price data needed for this study are not available at a daily frequency for many other asset classes, and stock transactions account for more than half of all the transactions in the dataset.

The unit of observation is an account-stock-day triple. Following Seru, Shumway, and Stoffman (2010), we count every purchase of a stock as the beginning of a new position, and a position ends on the date the investor first sells part or all of her holdings.³ Given that we have approximately 104,000 accounts that hold common stock, with an average of 3.5 stocks per

² “While individual stock accounting can potentially be a helpful way of thinking about the data, we emphasize that it is only a potential ingredient in an equilibrium model, and by no means a complete description of the facts. For one thing, we show that it underpredicts the correlation of stocks with each other, and argue that a model that combines individual stock accounting with broader forms of accounting is likely to be superior to a model that uses individual stock accounting alone.” (Barberis and Huang, 2001, page 1250)

³ We find similar results if we remove partial sales from our sample (i.e., not considering partial sales as the “death” of a position).

account over the 1,497 trading days in our sample, we begin with approximately 545 million potential observations. Following Ben-David and Hirshleifer (2012), we filter the raw dataset and make several simplifying assumptions. First, we include only securities that are identified as common shares and appear in CRSP. Because prices in the discount brokerage dataset are not adjusted for splits and dividends, we rely on CRSP factor adjustments to account for these issues. Second, we remove any account-stocks with negative commissions since they may indicate a reverse transaction. Third, account-stocks that include short sale transactions are removed to avoid any misrepresentation in the value-weighted average price (VWAP) of portfolio holdings. Fourth, we exclude positions for which we do not have information on the purchase price, which primarily arise when investors purchased stocks before the start of our sample period. Finally, since our primary question of interest is the effect of portfolio performance on investor behavior, we keep only account-days with at least two common stock holdings. After applying these filters and rules, we are left with a dataset of 110,554,055 (account, stock, day) observations. We report summary statistics in Panel A of Table 1.

[Insert Table 1 Here]

We have a similar brokerage dataset from China that ranges from 2000 to 2009.⁴ This dataset comes from a brokerage company that has multiple branches throughout China and serves approximately half a million investors. To make the analysis computationally feasible, we focus on a randomly selected 20% of the investors.⁵

⁴ Similar to the Barber and Odean data, the Chinese dataset contains three files: a trade file, a position file, and an investor demographic file. There are a few minor differences between the two samples. 1) Relative to the US data, the Chinese sample contains more limited information on investor demographics – only age and gender are available to researchers. 2) The position file is at the daily frequency in the Chinese dataset, so we do not have to build the account-stock-day triple observations based on the trade file like we do for the US data. 3) We do not have to remove short sale positions because short selling was not allowed in China during our sample period.

⁵ Specifically, we focus on the investors with an account number ending with either 0 or 5.

As with the US data, we restrict our analyses to common stocks and calculate holding period returns after adjusting for stock splits and dividends. Information on stock prices and distribution is taken from the China Stock Market and Accounting Research Database (CSMAR). After excluding positions for which we do not have information on the purchase price and excluding account-days where investors hold only one stock, the resulting dataset contains 97,000 unique investors and 84,793,767 (account, stock, day) observations. We report summary statistics for this sample in Panel B of Table 1. Note that the Chinese investors trade much more frequently than US investors – their daily selling probability ranges from 1.8% to 8.7%, depending on the status of the focal stock and the portfolio, while these numbers for the US investors are between 0.2% and 0.4%.

B. Main Methodology

Following Feng and Seasholes (2005), Seru, Shumway, and Stoffman (2010), and Barber and Odean (2013), we estimate the disposition effect using a hazard model which takes the following form:

$$h_{i,j}(t) = h_0(t) \exp\{\beta_0 + \beta_1 \text{Gain}_{i,j,t-1} + \epsilon_{i,j,t}\} \quad (1)$$

where observations occur at the account (i), stock (j), and date (t) level.⁶ For every account-stock-day, $h_{i,j}(t)$ is investor i 's probability of selling position j on day t conditional on not having sold prior to day t , and $h_0(t)$ is the baseline hazard. Additionally, *Gain* is a dummy variable equal to one if the stock's return since purchase (price/VWAP−1) is strictly positive and zero otherwise. With this structure, the hazard ratio, $\exp(\beta_1)$, measures the ratio of the probability of selling a winning position versus the probability of selling a losing position. Many previous studies show

⁶ We report our main results using the linear probability model in Tables A1 and A2 of the Internet Appendix.

that β_i is positive and statistically significant, or $\exp(\beta_1)$ is significantly greater than 1, suggesting that investors are more prone to liquidate winning positions than losing positions.

Our interest in this study is the relationship between the disposition effect and the performance of the investor's portfolio. We analyze this relationship by estimating the following equation:

$$h_{i,j}(t) = h_0(t) \exp\{\beta_0 + \beta_1 \text{Gain}_{i,j,t-1} + \beta_2 \text{Portfolio_Gain}_{i,t-1} + \beta_3 \text{Gain}_{i,j,t-1} \times \text{Portfolio_Gain}_{i,t-1} + \epsilon_{i,j,t}\}. \quad (2)$$

Our additional variable, *Portfolio Gain*, is a dummy indicating whether or not the investor's stock portfolio is at a gain or a loss. We compute this variable by first summing up the gains/losses (in dollars) of the investor's positions in all of her stocks as of the given day.⁷

Our main coefficient of interest in equation (2) is β_3 , the coefficient of the interaction term, which represents the ratio difference in disposition effects for paper gain portfolios and paper loss portfolios. In equation (2), $\exp(\beta_i)$ measures the disposition effect for paper loss portfolios, and $\exp(\beta_1 + \beta_3)$ measures the disposition effect for paper gain portfolios.

Compared to the linear probability model, which essentially estimates the disposition effect as the difference between the probability of selling winners and that of selling losers, the hazard model measures the disposition effect as the ratio of the two. This feature of the Cox (1972) proportional hazard model fits our research purpose particularly well: Because investors typically increase trading activity after positive portfolio performance (e.g., Gervais and Odean, 2001; Ben-David, Birru, and Prokopenya, 2018), the difference between the probability of selling winners

⁷ In Internet Appendix Table A3, we repeat our main analysis when the *Portfolio Gain* variable is defined without considering the performance of the stock associated with the given observation, i.e., when the portfolio gain is computed based on the performance of the *rest* of the investor's portfolio. The results are very similar.

and that of selling losers should mechanically change, while the ratio of the two should be immune to the change of turnover (Feng and Seasholes, 2005).

There are two ways to control for unobservable heterogeneity in the Cox proportional hazard model: fixed effects and stratification. The fixed effects model assumes the hazard rates between different groups are proportional, and it can be estimated by adding dummy variables to the right hand side of equation (2). However, the shortcoming is that it is difficult to incorporate a large number of fixed effects because the maximum likelihood estimator can suffer from the incidental parameters problem (Lancaster, 2000). The stratification method avoids the incidental parameters problem, and it relaxes the proportional hazard rate assumption of the fixed effect method and allows for different baseline hazards between the strata. In other words, with stratification, the baseline hazard function of $h_0(t)$ is allowed to vary across strata. Because of its flexibility, we use stratification to account for unobservable heterogeneity.⁸

III. The Portfolio-Driven Disposition Effect

A. Univariate Results

The phenomenon that we document in this paper, which we refer to as “the portfolio-driven disposition effect” (PDDE), can be illustrated with a simple figure.

[Insert Figure 1 Here]

We report this figure using the US (Panel A) and Chinese (Panel B) brokerage samples. This univariate view of the PDDE is strong in both samples.

⁸ There are two limitations of the hazard model. First, the hazard model does not allow for multiple dimensional stratification. For example, we cannot include investor strata and, at the same time, also date strata. In later analyses, we check the robustness of our results by specifying various strata, and we find that our results are robust. Second, we are unable to cluster the standard errors across multiple dimensions. We cluster by account, and we find that clustering by account gives more conservative t -values than clustering by stock or date.

Consider the probability that an investor sells one of her holdings. This is plotted in the portion of Figure 1 labeled “All Portfolios.” The disposition effect can be seen visually as the difference between the green (the probability of selling a gain) and the red (the probability of selling a loss) bars. The black bars (which represent all stocks) are included to show the weighted average. In Panel A, the unconditional probability of selling a given stock is approximately 0.24%. Adding the condition that a given stock’s return is positive (the green bar) increases the probability of an investor selling to 0.27%. The ratio (difference) in the probability of selling a gain versus a loss is approximately 1.42 (8 bps). In other words, an investor is approximately 42% ($0.27\%/0.19\% - 1$) more likely to sell a gain than a loss. This is the disposition effect.

To illustrate the PDDE, we reproduce these probabilities for two different scenarios: (1) the investor’s portfolio is at a gain (the portion labeled “> 0”), and (2) the investor’s portfolio is at a loss (the portion labeled “≤ 0”). The PDDE refers to the fact that the disposition effect is much stronger in the scenario where her portfolio is at a loss compared to when her portfolio is at a gain. In fact, in the US sample, the disposition effect ratio (difference) decreases to approximately 1.21 (4 bps) for gain portfolios. Conversely, the disposition effect grows substantially when the sample is restricted to observations where the portfolio is at a paper loss, resulting in a disposition effect ratio (difference) of approximately 2.14 (21 bps). These disposition effect ratios reveal that when an investor’s portfolio is at a paper loss (gain), she is 114% (21%) more likely to sell a gain than a loss.

As we mentioned earlier, the investors in our Chinese sample trade much more frequently: the unconditional probability of selling a given stock is approximately 3.62%, which is more than 15 times the probability in the US sample (Figure 1, Panel B). Nevertheless, the ratio (difference) in the probability of selling a gain versus a loss is approximately 3.73 (5.77%) for Chinese investors. In other words, an investor is approximately 273% ($7.88\%/2.11\% - 1$) more likely to sell a gain than a loss, indicating a strong disposition effect in the Chinese data. The PDDE is also strong in the Chinese sample: The disposition effect ratio (difference) decreases to approximately

1.73 (3.69%) for gain portfolios, and grows to approximately 3.56 (4.71%) for loss portfolios. These disposition effect ratios reveal that when an investor's portfolio is at a paper loss (gain), she is 256% (73%) more likely to sell a gain than a loss.

B. Baseline Regressions

We estimate equation (2) on the US (Chinese) sample described in Section II using a Cox hazard model and report the results in Table 2 Panel A (Panel B). Column 1 shows the baseline results with no stratification. Columns 2-4 add stratification by date, stock, and account, respectively. In column 5, we control for the V-shaped disposition effect (Ben-David and Hirshleifer, 2012) by including 52 return bracket indicators for the focal stock's return: $(-\infty, -50\%)$, ..., $[-4\%, -2\%)$, $[-2\%, 0)$, $[0, 2\%)$, $[2\%, 4\%)$, ..., $[50\%, +\infty)$. Additionally, one might be concerned that the variables *Gain* and *Portfolio Gain* are mechanically related; therefore in column 6, we consider an alternative definition for *Portfolio Gain* that excludes the focal stock when computing portfolio performance. We also consider other alternative definitions of *Portfolio Gain* such as the fraction of stocks in the portfolio that are at a gain. These results are similar to our main specification and are reported in Table A4 of the Internet Appendix.⁹

[Insert Table 2 Here]

Across all specifications, the coefficient on the interaction term (*Gain*Portfolio Gain*) ranges from -0.58 to -0.86 in the US data and -0.15 to -0.44 in the Chinese data. These coefficients indicate significant declines in the disposition effect when the portfolio is at a gain relative to when the portfolio is at a loss. For example, in our preferred specification with account stratification (column 4), the coefficient of *Gain* indicates a PGR/PLR ratio of 2.64 ($=e^{0.970}$) when

⁹ We also examine specifications with a standard set of controls following Ben-David and Hirshleifer (2012) and find very similar results. We report them in Internet Appendix Table A5.

the portfolio is at a loss. When the portfolio is at a gain, PGR/PLR decreases to 1.17 ($=e^{0.970-0.809}$). In the same specification of the Chinese data, PGR/PLR decreases from 2.11 for a winning portfolio to 1.36 for a losing portfolio. Moreover, these estimates are highly statistically significant, with t -stats all greater than 29. Taken together, these results suggest that the PDDE illustrated in Figure 1 is unlikely to be explained by unobservable investor, time, or stock characteristics that affect investors' propensity to sell shares of stock.¹⁰ Given that the account stratification gives a significantly better model fit (as reflected by the Pseudo R^2) than stock or date stratification, we use it as our preferred stratification and report most of the remaining analyses with this specification.

C. *The Magnitude of the Focal Stock and Portfolio Returns*

Next, we consider more continuous measures of the focal stock and the overall portfolio's performance. In Figure 2, we generate a heat map of hazard regression coefficients indicating relative selling probabilities as a function of the performance of the focal stock and the total portfolio. Specifically, we sort all the observations into 12-by-12 boxes by the focal stock's holding period returns and the total portfolio returns. Rows indicate different portfolio return brackets, and columns indicate different stock return brackets. The dependent variable is a dummy variable that equals 1 if there is sale (including partial sale) on day $t+1$, and 0 otherwise. We include dummy variables indicating each of the 144 combinations. The $(-\infty, -25\%]\times(-\infty, -25\%]$ group is the base case.¹¹ All the regressions are stratified by account. Areas with more (less) selling depict a darker shade of red (blue). The median is white.¹²

¹⁰ In Internet Appendix Table A6, we provide additional robustness by analyzing at the household level instead of the account level and find very similar results.

¹¹ In a linear regression, the coefficients on dummy variables that span a set of categories should be interpreted as a difference relative to the omitted group. Similarly, in hazard regressions, estimated coefficients on group dummies should be interpreted relative to the base case. For example, the coefficient on the $(-\infty, -25\%]\times(-25\%, -20\%]$ group is 0.10, indicating that this group's hazard is 10.5% (i.e., $\exp(0.10)-1$) higher than the base case.

¹² In Table A7 of the Internet Appendix, we provide heatmap examples of how a binary stock-level disposition effect (Panel A), a binary "portfolio-level disposition effect" whereby investors are more likely

[Insert Figure 2 Here]

The disposition effect can be observed in Figure 2 by noting that the right half of the two panels tends to be red, which indicates elevated selling activity, while the left half tends to be blue, which indicates reduced selling activity. Interestingly, the specific pattern of the disposition effect diverges across the two samples—There is a V-shaped selling schedule in the US sample (as documented by Ben-David and Hirshleifer, 2012), whereas the Chinese sample has a reverse V-shape with elevated selling activity near zero.

The presence of a V-shaped disposition effect in the US data and a reverse V-shape disposition effect in the Chinese data makes it less likely for us to find a PDDE in the US sample and more likely in the Chinese sample. To see why, consider the case where an investor exhibits a strong V-shape disposition effect (as in the US data). If her portfolio is at a gain, she is more likely to sell her individual stock gains because they are likely to be extreme gains, i.e., she will exhibit a stronger disposition effect when her portfolio is at a gain. This works *against* the PDDE. By the same logic, a reverse V-shape disposition effect works *for* the PDDE. Consistent with this reasoning, including V-shape controls in Table 2 strengthens the PDDE in the US data and weakens it in the Chinese data. Nevertheless, the PDDE remains strong in both samples.

As we move down each panel of Figure 2 (indicating improved portfolio performance), we see that the relative probability of selling losers (the left half of each panel) increases significantly. Conversely, the performance of the portfolio has a much weaker effect on the propensity to sell a winning stock (the right half of each panel). This pattern arises in both the US and Chinese samples, indicating that when we control for unobserved heterogeneity across accounts, the PDDE appears to be driven by the effect of portfolio return on the propensity to sell losing positions.

to sell a stock when the total portfolio is at a gain irrespective of the performance of the individual stock (Panel B), and a binary portfolio-driven disposition effect (Panel C) would materialize in isolation.

D. Subsample Analysis

We next examine how the PDDE varies with individual and portfolio characteristics. For each characteristic, we group all of the observations into subsamples, and we estimate the regression model from column 4 of Table 2 on various subsamples. We report all subsample tests in the US (Panel A) and Chinese (Panel B) brokerage samples. Because not all demographic information is available for all investors, the combination of all the subsamples is sometimes smaller than the whole sample. For investor characteristics, we study age, gender, and trading frequency. Investors are grouped into three age groups: 1 to 40, 41 to 55, and greater than 55. In the US, these three groups roughly correspond to the first quartile, quartiles 2-3, and quartile 4, respectively. In the Chinese sample, these three groups roughly correspond to quintiles 1-2, quintiles 3-4, and quintile 5, respectively.¹³ Trading frequency is calculated as the unconditional selling propensity of an investor over the whole sample period. We sort all the investors into two trading frequency groups: those above and below the median. For portfolio characteristics, we test the impact of holding period on the PDDE. To employ the hazard model described in Section IIB, we must include all observations from the start of the holding period until a maximum number of days. Thus, we report two variations of the maximum holding period, 20 days and 250 days.

[Insert Table 3 Here]

Several observations emerge from Table 3. First, we find that the moderating effect of portfolio gain is similar across the different age and gender groups but is slightly larger for high (low) trading frequency investors in the US (Chinese) sample.¹⁴ Second, the PDDE is larger for

¹³ In the Chinese sample, age 40 is the 39th percentile, and age 55 is the 79th percentile.

¹⁴ The t-stat on Female is reduced in the US sample because they make up only about 10% of the observations. The gender distribution is significantly more balanced in the Chinese sample, with females representing roughly 54% of the observations. Regardless, the economic magnitudes of the interaction coefficients are similar in both samples.

longer holding periods in both the US and Chinese samples. Finally, the moderating role of portfolio gain on the disposition effect remains strong across all subsamples, indicated by the significantly negative interaction coefficients ranging from -0.501 to -0.849 (t-stats range from -9.64 to -41.64) in the US sample and -0.274 to -0.606 (t-stats range from -47.66 to -91.48) in the Chinese sample.

E. Matching Analysis

In an ideal experiment, we would compare identical positions in a particular stock owned by identical investors, with the only difference being the investors' portfolio performance. By identical positions, we mean that both investors own the same stock, and they purchased the stock on the same day and at the same price. By identical investors, we mean investors who would make the same decisions when facing any economic scenario. Because of our large sample, we have identical positions; however, this ideal experiment is not feasible because we do not have identical investors. In this matching analysis, we approximate the ideal experiment by comparing identical positions owned by different investors. Specifically, we stratify by positions built on the same day and of the same stock in the regressions. By doing this matching, we keep the stock and purchase date the same and focus on the portfolio return variation across investors. We also only keep the observations where there are at least two investors within the same strata. The number of observations is 52.2% of the entire sample. We find that the coefficient of the interaction term is negative and highly significant whether we stratify by account (-0.831 , t-stat -38.90) or by stock*purchase date (-0.588 , t-stat -34.13). We report these results in columns 1 and 2 of Table 4.

[Insert Table 4 Here]

With the stratification by positions built on the same day and of the same stock, there is not much variation in purchase price within a stratum because we consider purchases of the same stock on the same day. Nevertheless, in the second specification, we further require the exact same purchase price in the stratification. Relative to the first specification, this filter reduces the sample by around half, and yet the PDDE effect is similar in magnitude, with interaction coefficients of $(-0.838, t\text{-stat } -30.26)$ and $(-0.550, t\text{-stat } -23.30)$ in columns 3 and 4, respectively. In the third specification, columns 5 and 6, we further exclude the positions that were constructed with multiple purchases, and we continue to find a strong PDDE.¹⁵

We repeat the above analyses for the Chinese sample.¹⁶ The coefficient of the interaction term remains negative and highly significant.¹⁷ The results are broadly similar across the six specifications. The relative magnitude of the *Gain* and *Gain*Portfolio Gain* terms are similar to the baseline estimation in Table 2, which suggests that the PDDE is not driven by unobserved stock-level characteristics that are correlated with the portfolio's gain/loss status.

F. Experimental Settings

In the prior section, we compare the selling decision for the same stocks at the same point in time across investors whose portfolios are at a gain and a loss. While this is as close to ideal in the archival setting as possible, we can come even closer to the ideal setting in an actual experiment where we can do several things which are not possible in the archival setting: (i) have the same trader on the same day be faced with a portfolio gain and a portfolio loss and observe her differential behavior across these conditions, (ii) inform each trader of the stock data-

¹⁵ In other words, in this last specification, we only consider (investor, stock, purchase date) triples such that the investor liquidates some of her position in the stock before purchasing any more shares of the stock.

¹⁶ During our sample period, in the US, the tick size was one eighth of a dollar. In China, the tick size was one cent RMB. To be consistent, in the Chinese data, we round the purchase prices to the nearest eighth and require that the rounded purchase price be the same.

¹⁷ If we do not round the purchase price, the coefficient of the interaction term is -0.290 ($t\text{-stat } -4.89$) and -0.313 ($t\text{-stat } -5.54$) in column 4 and 6, respectively. The number of observations is 2,525,960 and 2,166,353, respectively. Both are significantly smaller than the analysis with the rounded price.

generating process to minimize differences in information across traders, and (iii) control the display of information to traders.

We adopt a trading game that is similar to Weber and Camerer (1998). We recruited subjects from two pools: undergraduate students from a Chinese university and volunteer workers (“MTurkers”) from Amazon’s Mechanical Turk (“MTurk”). We required subjects to pass a quiz administered prior to the trading game, and we collected trading data from 766 and 2,377 subjects in the Chinese university and MTurk samples.¹⁸ There are four fictitious stocks, labeled A, B, C, and D, whose prices randomly evolve. Each trading game lasts 12 rounds. In the first two rounds, subjects just observe the stock prices and are not able to trade. In Rounds 3 through 12, subjects can use their experimental cash to purchase shares of the stocks by using a “Buy” button, and if they own shares of a stock, they can sell shares of it using a “Sell” button. Each subject plays this 12-round stock trading game a total of 4 times. Subjects are not allowed to borrow cash or short sell.

Stock prices evolve each round according to the following two independent processes. First, it is randomly determined whether a stock’s price will increase or decrease. One of the stocks is randomly chosen to have a 65% chance each round of having its price increase, while the other three stocks are randomly assigned the probabilities 55%, 45%, and 35%. The probabilities of increases or decreases are different for each stock and do not change between rounds within a game. Across games, however, the probabilities are randomly reassigned. Second, the magnitude of the price increase or decrease is determined. Prices either change by \$1, \$3, or \$5, each with equal probability. The magnitude of the price changes is independent of whether the price is increasing or decreasing.

¹⁸ We administered the quiz to 1,295 Chinese university students and 9,600 people from MTurk. Of these, a total of 811 (62.6%) and 2,537 (26.4%) Chinese university and MTurk subjects passed the quiz and played the trading game. Due to a server failure, we do not have complete trading data for some of these subjects, leaving us with a final sample of 766 and 2,377 subjects in the Chinese university and MTurk samples. Thus, our pooled sample of MTurk and Chinese university students consists of 3,143 subjects. The end of the Internet Appendix includes the pre-quiz introductory information, quiz questions, and experimental treatment descriptions.

Each round, subjects are shown a graph of each stock's price evolution up to that round.¹⁹ Each game, subjects are endowed with 1,000 units of experimental cash that they can use to trade the fictitious stocks, and their compensation depends on the total value of their assets (stock plus experimental cash) at the end of the experiment. Specifically, subjects are paid a 1 RMB (\$0.25) show-up fee in the Chinese university (MTurk) sample plus a bonus that is based on their performance during one of the four trading games, which is randomly chosen. The average pay for the subjects was 20.93 RMB and \$3.25 in the Chinese and MTurk samples, respectively.

We begin by estimating our baseline hazard regression model in Table 5. In columns 1-3, we stratify by subject, and in columns 4-6, we also add controls for the V-shaped disposition effect (Ben-David and Hirshleifer, 2012) by including the same return bracket fixed effects from column 5 of Table 2. We report the estimates for the Chinese sample in columns 1 and 4, the MTurk sample in columns 2 and 5, and the pooled sample consisting of all subjects in columns 3 and 6.

[Insert Table 5 Here]

The positive coefficients on *Gain* in columns 1-3 demonstrate a strong disposition effect in our experimental settings. As in the previous tables, we look for a PDDE by examining the interaction of *Gain* and *Portfolio Gain*. In each specification, we find that the coefficient of the interaction is negative and statistically significant, consistent with subjects exhibiting a PDDE. For example, consider the coefficient of 0.699 on *Gain* and -0.420 on $Gain * Portfolio\ Gain$ in column 1. The coefficient on *Gain* indicates a PGR/PLR ratio of 2.01 ($=e^{0.699}$) when the portfolio is at a loss. When the portfolio is at a gain, PGR/PLR decreases to 1.32 ($=e^{0.699-0.420}$). Additionally, we find that in the most controlled tests with return bracket FEs the interaction coefficient is -0.362 (t-stat -6.93) in the Chinese experimental sample, -0.421 (t-stat -10.71) in the MTurk

¹⁹ We discuss additional details about the display of portfolio performance in Internet Appendix Section A2.

sample, and -0.400 (t-stat -12.74) in the pooled sample. Together, these results provide evidence that the PDDE holds in this well-controlled environment where we observe the same person on the same day exposed to both a portfolio gain and loss and can observe her differential tendency to exhibit the disposition effect.

G. Aggregate Implications

The PDDE has natural aggregate implications. Although conventional wisdom suggests that the disposition effect is idiosyncratic and specific to each individual investor, the moderating role of portfolio performance can generate aggregate and cyclical effects because the performance of individual portfolios is commonly driven by the overall market. Therefore, one implication of the PDDE is that all investors tend to exhibit the disposition effect around similar points in time, or in other words, there should be a “disposition effect comovement.”

To test this prediction, we calculate the level of the disposition effect across different investor groups quarter by quarter in both the US and Chinese samples. We stratify investors by gender, age, portfolio size (into ten equal-sized groups), and in the fourth test, randomly (into ten equal-sized groups). For each investor group in each quarter, we estimate the average disposition effect by running equation (1) using the investor-stock-day observations. Internet Appendix Figure A1 presents the results. The y-axis is the value of the disposition effect, and the x-axis is quarter. We see that investors with different gender, age, portfolio size, and other characteristics comove very closely over time in the level of the disposition effect.

Moreover, the PDDE also predicts that the time-series variation of the disposition effect should be related to past market performance: After a bull market, most investors' portfolios will be at a gain, and therefore, the PDDE implies that they should exhibit a weaker disposition effect. In contrast, a bear market should lead to portfolio losses for most investors, and thus, a strong disposition effect among investors. In Internet Appendix Table A10, we examine the relation between the quarterly average disposition effect across all the investors and various horizons of

past market returns in a univariate regression framework. We find that the quarterly average disposition effect is negatively correlated with past market returns at almost all horizons in both samples.²⁰ Moreover, when we compare the correlation across different horizons, we find that the negative correlation between the disposition effect and past market return peaks at eight quarters in the US sample and at three quarters in the Chinese sample. Interestingly, these horizons closely match investors' average holding periods in the two samples.²¹ These coefficients are economically sizable. For example, in the US, a one standard deviation (13.1%) change in the cumulative market return over the past eight quarters ($R_{t-8,t-1}$) is associated with a 0.083 decrease of the disposition effect, which is 18% of the average of the quarterly disposition effect (0.455). In China, a one standard deviation (41.6%) change in $R_{t-3,t-1}$ (the cumulative market return over the past three quarters) is associated with a 0.112 decrease of the disposition effect, which is 17% of the average of the quarterly disposition effect (0.667).

[Insert Figure 3 Here]

Figure 3 presents the time series of the average disposition effect and the past market returns for the US and Chinese samples, with past market returns measured over the past eight (three) quarters for the US (China). The negative correlation between the disposition effect and past market return is evident.²² Thus, we document a systematic and cyclical component in one of the most robust behavioral patterns, the disposition effect.

²⁰ Bernard, Loos, and Weber (2022) find similar patterns using individual investor trading data from Germany.

²¹ According to World Bank, the average market turnover during our sample periods is 65% (implying an average holding period of six quarters) in the United States and 195% (implying an average holding period of two quarters) in China.

²² We also investigate whether the PDDE is different in Boom and Bust periods in Internet Appendix Table A11. We find no difference in the PDDE between Boom and Bust periods in the US and a slightly stronger PDDE during Bust periods in China.

IV. Relationship to Prior Research

In this section, we examine whether the PDDE we document is simply a manifestation of prior empirical research.

A. *The Rank Effect*

We first test whether extreme stocks drive the PDDE. Hartzmark (2015) finds that retail and mutual fund investors are more likely to sell their best and worst performing stocks. Intuitively, these extreme stocks grab the investor's attention and, as a result, are sold more often. In our setting, the attention-grabbing hypothesis could predict some of our results, but not others. For example, if an investor has one stock that is a winner and the rest losers, then this stock is very likely to be sold under both the attention-grabbing hypothesis (it is an extreme stock) and the PDDE (investors are very likely to sell their winners when the portfolio is at a loss). However, if an investor has one stock that is a loser and the rest winners, this stock is very likely to be sold under the attention-grabbing hypothesis (because it is an extreme stock) but not the PDDE (because losers are nearly as likely to be sold as winners are when the remaining portfolio is at a gain).

[Insert Table 6 Here]

Nevertheless, in Table 6 we evaluate how the rank effect relates to our empirical results. Specifically, in column 1, we add indicator variables for the best performed and the worst performed stocks in an investor's portfolio. In column 2, we add indicator variables for each of the 15 stocks with the best performance and the 15 stocks with the worst performance in an investor's portfolio, following Hartzmark (2015). The interaction coefficient is very similar to the baseline regression in column 4 of Table 2 both in terms of statistical significance and economic

magnitude for the US (Panel A) and Chinese (Panel B) samples. These results suggest that the rank effect (Hartzmark, 2015) does not explain the PDDE.

B. Tax Incentives

Because Odean (1998) finds that the disposition effect is time-varying due to tax motivations, we consider the possibility that the PDDE could be explained by such considerations. Thus, we conduct our analyses separately within US tax-exempt accounts. Columns 3 and 4 of Table 6 Panel A report the regression results splitting the US sample by tax-exempt and taxable accounts, respectively. Although the PDDE is slightly stronger for taxable accounts, the PDDE is strong among tax-exempt accounts with an interaction coefficient of -0.700 (t-stat -21.93). Furthermore, tax-loss selling cannot explain the PDDE in the Chinese sample, where capital gains are not taxed. These findings suggest that tax considerations do not explain the PDDE.

C. Portfolio Rebalancing

Although Odean (1998) provides evidence that portfolio rebalancing does not explain the disposition effect, it is possible that portfolio rebalancing causes the PDDE. For example, suppose all but one of an investor's stocks are at a loss. It is likely that the lone stock that is trading at a gain comprises a disproportionately large percentage of the investor's portfolio due to its gains and the rest of the stocks' losses. The investor might therefore want to liquidate some of her holdings in the stock that is at a gain in order to rebalance her portfolio. According to this explanation, we should expect investors to *partially* (not completely) liquidate their positions in the stock that is at a gain when the rest of the portfolio is at a loss. That is, we should expect the PDDE to disappear when we restrict attention to *complete* liquidations of stocks.

To test this, we adjust our specification to use a full liquidation dummy as the dependent variable, thus eliminating any variation from partial sales. In column 5 of Panel A and column 3 of Panel B of Table 6, we report the full liquidation results. For the US, we see that the interaction

coefficient of -0.915 (t-stat -49.84) is still negative and significant well below the 1% level and nearly offsets the magnitude of the *Gain* coefficient. For the Chinese sample, the interaction coefficient is -0.597 (t-stat -115.57) and is about two-thirds of the *Gain* coefficient. Thus, portfolio rebalancing is an unlikely explanation for the PDDE.

D. Investor Heterogeneity of the Disposition Effect

Another possibility is that the PDDE is explained by heterogeneity of the disposition effect across investors, which may stem from various sources such as investor IQ (Grinblatt, Keloharju, and Linnainmaa, 2012).²³ Specifically, people who have a strong disposition effect tend to sell winners in their portfolio and keep losers. Therefore, individuals with a strong disposition effect are more likely to have a (paper) portfolio loss compared to individuals with a weak disposition effect, a pattern that might confound our findings.

To address this concern, we employ the following two approaches. First, when calculating *Portfolio Gain*, instead of restricting attention to the paper gain/loss of currently held positions, we add back previously realized gains/losses in the past one year as if they were not realized.²⁴ Under this construction, the variation of *Portfolio Gain* is mainly driven by the performance of securities in the portfolio, and is unrelated to whether these positions are realized or not (and thus the investor's tendency to exhibit a disposition effect). Table 7 columns 1 and 3 report results based on *Portfolio Gain* constructed this way.

[Insert Table 7 Here]

²³ We discuss the relation between investor sophistication and the PDDE in more detail in Internet Appendix Section A1. Internet Appendix Table A12 provides evidence that the estimation of PDDE does not vary across investor subsamples based on income levels and occupations.

²⁴ In Table A13 of the Internet Appendix, we show results using different horizons for adding realized gains/losses and find that the PDDE persists across all specifications.

Second, we control for heterogeneity in the disposition effect across investors by stratifying the baseline hazard function at the account-gain level in columns 2 and 3 instead of the account level as in column 1. This specification is designed to control for the variation in the disposition effect across investors. That is, for each account, we explicitly allow for his/her propensity to sell (the baseline hazard function) to differ across winning and losing positions.²⁵ As shown in Table 7 columns 1-3, under the alternative definition of *Portfolio Gain* and/or the more saturated stratification, the coefficient estimate on the interaction term between *Gain* and *Portfolio Gain* remains significantly negative. Note that since we have subsumed the baseline, account-specific disposition effect under these specifications, the magnitude of the coefficient on the interaction term is not directly comparable across columns, as it reflects a proportional change relative to the baseline case. However, within each column, the magnitude of the coefficient on the interaction term is similar to that on *Portfolio Gain* in both the US and Chinese samples. This observation suggests that investor heterogeneity in the disposition effect can only explain part of the PDDE.²⁶

V. Multiple Frames

The prior section rejected explanations where the PDDE is a byproduct of various possible mechanisms, such as the rank effect or investor heterogeneity of the disposition effect. In this section, we present evidence that investors have multiple frames which combine to generate the PDDE.²⁷

²⁵ Note that the estimation from this specification is likely to be conservative. Controlling for the individual heterogeneity in the disposition effect will mechanically lead to a negative autocorrelation of the disposition effect. To the extent that *Portfolio Gain* at t is an inverse function of the disposition effect before t , the account-gain stratification tends to underestimate the PDDE effect (e.g., a less negative coefficient on *Gain* $\times$ *Portfolio Gain*). In Internet Appendix Table A14, we show that the results in the experimental samples are also robust to subject-gain stratification.

²⁶ Additionally, investor heterogeneity in the disposition effect cannot explain the correlation between the disposition effect and past market returns, or the comovement in the disposition effect across different investor types, which we document in Section III.G.

²⁷ The idea that investors have multiple frames is consistent with the contemporaneous theoretical work of Dai, Qin, and Wang (2023).

Prior research on the disposition effect has established narrow framing at the stock level, among stocks within a portfolio (Hartzmark, 2015), and even framing across trades (Frydman, Hartzmark, and Solomon, 2018). Here we present evidence that investors simultaneously use two separate frames – one at the stock level and one at the portfolio level - when making decisions, resulting in a PDDE. There are several ways that multiple frames could generate the PDDE. For example, investors might engage in hedonic mental accounting (Thaler, 1985), which posits that people frame their decisions in the way that makes them feel best. Specifically, by framing the sale of a losing stock as a liquidation of (part of) the larger portfolio, the investor can mentally account for the liquidation of a loss as a gain, but this is only possible when the portfolio is at a gain. Another possibility is cognitive dissonance, which has been proposed as a mechanism for the disposition effect (Chang, Solomon, and Westerfield, 2016). An investor who simultaneously frames at both the individual stock level and the portfolio level should be especially prone to exhibit the disposition effect when her portfolio is at a loss: in this case, both frames (stock and portfolio) suggest that the investor made bad decisions, and liquidating a loss in this scenario should be particularly difficult to reconcile with their self-image of being someone who makes good decisions.²⁸ Conversely, if the investor's portfolio is at a gain, she can still convince herself she is a good trader by paying attention to the portfolio-level frame, which suggests she is a good trader who makes good decisions.

In order to provide direct evidence of an additional frame at the portfolio level, we use the fact that a focal asset's membership in a portfolio will be determined based on how similar the

²⁸ As Chang, Solomon, and Westerfield (2016) explain: “[I]nvestors feel a cognitive dissonance discomfort when faced with losses—there is a disconnect between the belief that the investor makes good decisions and the fact that the investor has now lost money on the position. While all losses cause such dissonance, realized losses create a greater level of discomfort than paper losses: when the loss exists only on paper the investor is able to partly resolve the dissonance by convincing themselves that the loss is only a temporary setback. This reduces the blow to their self-image of being someone who makes good decisions. When the loss is realized, it becomes permanent, which makes it harder for the investor to avoid the view that buying the share may have been a mistake. Cognitive dissonance provides the basis for an overall reluctance to realize losses, and thus generates a wedge relative to the investor's propensity to realize gains (where no such dissonance discomfort exists).”

other assets in the portfolio are. Similarity has been extensively documented as a defining characteristic of how individuals create mental accounts (Evers, Imas, and Kang, 2022; Goldstone, 1994; Kruschke, 1992; Nosofsky, 1986). If a portfolio-level frame drives the PDDE, similarity has a direct prediction: assets which are most similar to the focal stock (i.e., most likely to be in the same mental account as the focal stock) should have the greatest contribution to the PDDE. With this in mind, we perform two tests based on similarity.

A. Evidence from Account Similarity

First, we exploit the fact that a single household can have multiple accounts from the US discount brokerage. Two stocks in the same account are more likely to be considered in the same portfolio than two stocks in different accounts, even though all stocks in the multiple accounts contribute to the household's wealth. We decompose the household return into those stocks in the same account as the focal stock and those stocks held in different accounts in the same household as the focal stock by revising equation (2) as follows:

$$\begin{aligned}
 h_{i,j}(t) = & h_0(t) \exp \{ \beta_0 + \beta_1 \text{Gain}_{i,j,t-1} + \beta_2 \text{AccountPortfolioRet}_{i,t-1}^{all-j} \\
 & + \beta_3 \text{Gain}_{i,j,t-1} \times \text{AccountPortfolioRet}_{i,t-1}^{all-j} + \beta_4 \text{HouseholdPortfolioRet}_{i,t-1}^{all-Account} \\
 & + \beta_5 \text{Gain}_{i,j,t-1} \times \text{HouseholdPortfolioRet}_{i,t-1}^{all-Account} + \varepsilon_{i,j,t} \}
 \end{aligned} \tag{3}$$

where $\text{AccountPortfolioRet}_{i,t-1}^{all-j}$ is the return from the other stocks in the same account as focal stock j . $\text{HouseholdPortfolioRet}_{i,t-1}^{all-Account}$ is the return from all household stocks excluding the stocks in the same account as focal stock j . We define return variables as continuous variables rather than dummies in order to decompose the remaining returns of a household into $\text{AccountPortfolioRet}_{i,t-1}^{all-j}$ and $\text{HouseholdPortfolioRet}_{i,t-1}^{all-Account}$. Thus β_3 measures the moderating effect of stock portfolio performance on the disposition effect from performance in

the same account as the focal stock, while β_5 captures the moderating effect of household gains that originate from other accounts. To decompose household return, we sum the capital gains within each account and normalize them by the sum of the cost basis of the entire household. Therefore, the sum of the account returns within a household is equal to the overall household return. Moreover, capital gains accumulated across accounts are comparable because both β_3 and β_5 capture the marginal effect of the same dollar amount of household returns. For these tests, we require a household to have at least two accounts. The final sample has approximately 50 million account-stock-date observations.²⁹

[Insert Table 8 Here]

In Table 8, we decompose the household return into those stocks in the same account as the focal stock and those stocks held in different accounts. Consistent with the prediction of a portfolio-level frame, the PDDE moderates following dissimilarity. Specifically, column 1 shows that stocks held by the same account as the focal stock generate a PDDE which is 21% ($90.0\%/74.4\% - 1$) larger than that of stocks held in the same household but a different account than the focal stock.³⁰ Moreover, the p-value testing the difference between β_3 and β_5 is less than 1%. When restricting to one-adult households, the effect of dissimilarity grows: other holdings in the focal account have a 28% ($87.6\%/68.5\% - 1$) larger PDDE compared to that of holdings in different accounts of the same household.³¹ Additionally, we reject the null that $\beta_3 = \beta_5$ at the 1% level.

²⁹ In Appendix Table A15, we verify that the PDDE holds when a household's portfolio is constructed using all assets in all accounts within the household.

³⁰ As shown in column 1, a one-unit increase in portfolio return generated by other stocks in the same account leads to a 90.0% ($1 - e^{\beta_3} = 1 - e^{-2.302}$) decrease in the disposition effect, while this number for a portfolio return generated by stocks in other accounts is 74.4% ($1 - e^{\beta_5} = 1 - e^{-1.364}$).

³¹ For one-adult households, as shown in column 2, a one-unit increase in portfolio return generated by other stocks in the same account leads to a 87.6% ($1 - e^{\beta_3} = 1 - e^{-2.088}$) decrease in the disposition effect, while this number for return generated by stocks in other accounts is 68.5% ($1 - e^{\beta_5} = 1 - e^{-1.156}$).

B. Evidence from Asset Similarity

In the prior section, we measured the similarity of stocks across accounts within the same household. Here we exploit the characteristics of the individual assets within a single brokerage account. For example, two US common stocks are more similar to each other than one US common stock and one mutual fund in the same brokerage account.

Specifically, we explore whether portfolio gains from different asset classes affect the disposition effect of US stocks in the same way.³² To conduct the asset-class similarity analysis, we make some necessary modifications to our baseline model. We revise the model in equation (2) to the following:

$$\begin{aligned}
 h_{i,j}(m) = & h_0(m) \exp\{\beta_0 + \beta_1 \text{Gain}_{i,j,m-1} + \beta_2 \text{PortfolioRet}_{i,m-1}^{\text{CommonStock}-j} \\
 & + \beta_3 \text{Gain}_{i,j,m-1} \times \text{PortfolioRet}_{i,m-1}^{\text{CommonStock}-j} + \beta_4 \text{PortfolioRet}_{i,m-1}^{\text{Category}} \\
 & + \beta_5 \text{Gain}_{i,j,m-1} \times \text{PortfolioRet}_{i,m-1}^{\text{Category}} + \varepsilon_{i,j,m}\}
 \end{aligned} \tag{4}$$

where m indicates time and is now monthly because daily prices are not available for many asset classes during the sample period. $\text{PortfolioRet}_{i,m-1}^{\text{CommonStock}-j}$ is the account i 's portfolio return from the US common stock, excluding the focal stock j .³³ $\text{PortfolioRet}_{i,m-1}^{\text{Category}}$ is account i 's portfolio return from another category of asset classes. As in equation (2), β_3 measures the moderating effect of US common stock portfolio performance on the disposition effect, and β_5 captures the moderating effect of portfolio return from a different asset-class category. We sort investors' assets into US common stocks, foreign stocks (mainly Canadian stocks and ADRs),

³² We examine these predictions using the US data because we do not have trading data on other securities in the Chinese dataset. To this point, we have only analyzed the US common stock holdings of investors in the US sample, but now we expand their portfolios to analyze the influence of returns from other asset classes.

³³ $\text{PortfolioRet}_{i,m-1}^{\text{CommonStock}-j}$ consists of the same set of stocks as $\text{AccountPortfolioRet}_{i,t-1}^{\text{all}-j}$.

open-end mutual funds, options, and other stock-type securities (mainly closed-end mutual funds, master limited partnerships, and preferred stocks).³⁴ Consistent with the analysis in the prior section, we calculate all asset-class returns by summing the capital gains within that asset class and normalizing them by the cost basis of the entire portfolio so that the sum of all asset-class returns is equal to the overall portfolio return. We follow Chang, Solomon, and Westerfield (2016) in calculating gains and losses for securities in each asset class at a monthly frequency.³⁵ To ensure a fair comparison, we use monthly price information in the positions file for securities in all asset classes, including US common stocks. This procedure yields 3.75 million account-stock-month observations. To conduct our analysis, we require that an investor holds at least two common stocks as well as a security in any of the other asset classes in a given month. This final sample consists of approximately 0.7 million observations.³⁶ Internet Appendix Table A17 verifies that the PDDE holds in this sample.

[Insert Table 9 Here]

In Table 9, we decompose the overall portfolio return and compare capital gains from US common stocks against those from the other four categories one by one. We observe that the moderating effect of the US common stock portfolio (excluding the focal stock) is highly statistically significant across all of the specifications. Furthermore, the moderating effect of

³⁴ See Internet Appendix Table A16 for a detailed description of each asset category.

³⁵ Specifically, we use both the trade file and the position file, and we delete the observations where the holdings in the positions file cannot be matched with those inferred by trading records in the trades file. Purchase price is calculated as the volume-weighted average prices using the trades file. We evaluate the gains and losses for each security at the end of each month. We obtain a monthly snapshot of security prices as long as a security is held by at least one investor in the dataset. We exclude money market funds because many have a price that is fixed and therefore have very few observable gains and losses.

³⁶ We only analyze the disposition effect among US common stocks (i.e., other asset classes are never the focal asset) for two reasons. First, Chang, Solomon, and Westerfield (2016) show that in many delegated assets (such as mutual funds), the disposition effect actually reverses or disappears. Second, our regression design would require that the investor hold at least two securities in the given asset class in addition to securities in other asset classes, which greatly reduces the number of observations available.

capital gains generated by other asset categories is significantly smaller than that of US common stocks, as indicated by the smaller magnitude of β_5 compared to β_3 . Economically speaking, the moderating effect of one unit of capital gains generated by other US common stocks in the portfolio is 25% ($83.0\%/66.5\% - 1$) larger than that of foreign stocks and 24% ($84.9\%/68.5\% - 1$) larger than that of other stock-type securities.³⁷ Additionally, we reject the null that $\beta_3 = \beta_5$ at the 1% level. The moderating effect of other US common stock is 3 to 4 times as large as those of mutual funds and options; in fact, the estimations of the latter two are not statistically significant. These findings provide additional evidence of a portfolio-level frame: when assets are similar and therefore more likely to be in the same mental account as the focal stock, those assets generate a stronger PDDE than assets that are dissimilar.

Together, the PDDE is increasing in both asset and account similarity. These results provide evidence that investors not only frame at a stock level but also at a portfolio level, and the combination of these two frames generates the PDDE.

VI. Conclusion

The disposition effect is a stock-level phenomenon. But individuals rarely hold single stocks; they often hold portfolios. The purpose of this paper has been to answer the question: does the stock-level disposition effect depend on the portfolios they hold? We find a consistent answer among four independent settings: 78,000 US households in a large discount brokerage, 97,000 investors in a Chinese brokerage, 2,300 US participants in an Amazon Mechanical Turk (“MTurk”) trading-game experiment, and 800 experimental participants at a Chinese university.

³⁷ Column 1 shows that, a one-unit increase in portfolio return generated by other US common stocks leads to an 83.0% ($1 - e^{\beta_3} = 1 - e^{-1.770}$) decrease in the disposition effect, while this number for a portfolio return generated by foreign stocks is 66.5% ($1 - e^{\beta_5} = 1 - e^{-1.093}$). Column 2 shows that, a one-unit increase in portfolio return generated by other US common stocks leads to a 84.9% ($1 - e^{\beta_3} = 1 - e^{-1.888}$) decrease in the disposition effect, while this number for a portfolio return generated by other stock-type securities is 68.5% ($1 - e^{\beta_5} = 1 - e^{-1.155}$).

In each of these settings, an investor's disposition effect is large when her portfolio is at a loss and significantly smaller when it is at a gain.

This portfolio-driven disposition effect is robust to a variety of controls and does not seem to be a repackaging of previously documented research concerning the disposition effect. However, we find direct evidence that the PDDE is a byproduct of investors using an additional frame – at the portfolio level – when making investment decisions. In doing so, the PDDE contributes to our understanding of how people frame financial decisions. Originally, researchers assumed that investors use fairly static and fixed frames, but recent research suggests that framing is more fluid and nuanced. Sometimes, individuals engage in relative evaluation within a portfolio (Hartzmark, 2015), and they sometimes frame across trades (Frydman, Hartzmark, and Solomon, 2018). Our evidence suggests investors frame at multiple levels—the stock level and portfolio level— when making trading decisions.

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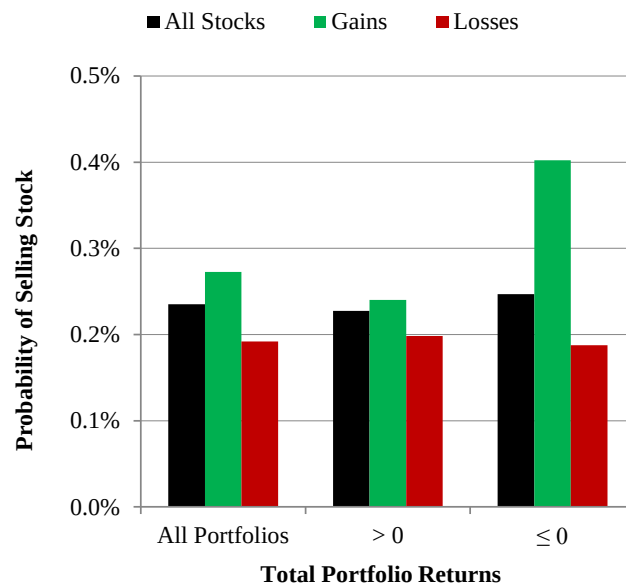
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Figure 1. Probability of selling a stock based on its return and the return of the portfolio

We report the probability of selling a stock (including partial sales) based on the stock's performance (gain versus loss) from the date the investor purchased the stock and the performance of the investor's portfolio of stocks. In Panel A (Panel B), we report the results using the US (Chinese) brokerage sample described in Section II. The US sample has 110,554,055 observations (53% stock gains, 47% stock losses; 61% total portfolio gains, 39% total portfolio losses). The Chinese sample has 84,796,020 observations (26% stock gains, 74% stock losses; 78% total portfolio gains, 22% total portfolio losses). We define gains (green bars) as strictly greater than zero while losses (red bars) include zeros.

Panel A. US



Panel B. China

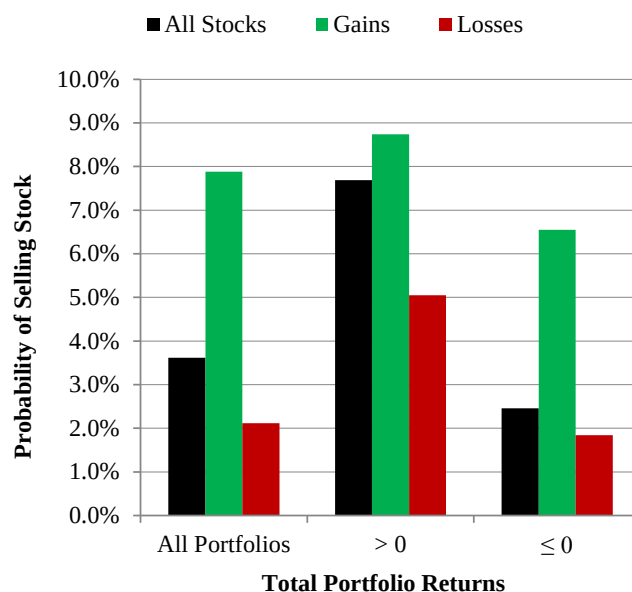


Figure 2. Non-binary measures of the focal stock and portfolio returns

We sort the sample into 12-by-12 boxes by the focal stock's holding period return and the total portfolio return, defining an indicator variable for each box. We report the Cox hazard regression coefficients for these 144 indicator variables in this table. Rows indicate different portfolio return brackets, and columns indicate different focal stock return brackets. The dependent variable is a dummy variable that equals 1 if there is sale (including partial sale) on day $t+1$, and 0 otherwise. The $(-\infty, -25\%]\times(-\infty, -25\%]$ group is the base case. Regressions are stratified by account. Areas with more (less) selling are depicted by a darker shade of red (blue). The median is white.

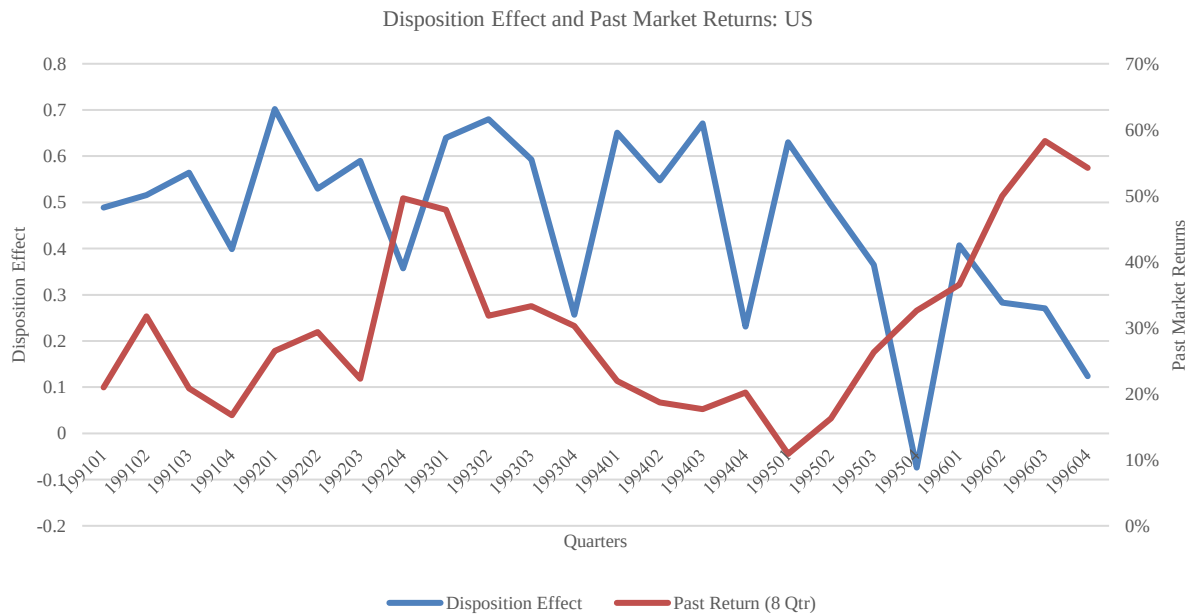
Panel A. US		Focal stock return											
		(-∞,-25%]	(-25%,-20%]	(-20%,-15%]	(-15%,-10%]	(-10%,-5%]	(-5%,0]	(0,5%]	(5%,10%]	(10%,15%]	(15%,20%]	(20%,25%]	(25%,+∞)
Portfolio return	(-∞,-25%]	0.00	0.10	0.19	0.18	0.18	0.34	0.90	1.24	1.40	1.46	1.70	1.54
	(-25%,-20%]	0.35	-0.24	0.07	0.21	0.22	0.36	0.91	1.38	1.49	1.71	1.78	1.74
	(-20%,-15%]	0.43	0.27	-0.11	0.27	0.21	0.41	0.96	1.32	1.66	1.70	1.65	1.77
	(-15%,-10%]	0.58	0.42	0.37	0.02	0.28	0.39	0.99	1.37	1.47	1.67	1.59	1.73
	(-10%,-5%]	0.70	0.62	0.49	0.43	0.15	0.42	1.01	1.34	1.51	1.71	1.74	1.77
	(-5%,0]	1.02	0.89	0.82	0.72	0.56	0.38	0.95	1.32	1.49	1.62	1.66	1.75
	(0,5%]	1.22	1.09	1.10	0.96	0.84	0.66	0.83	1.26	1.43	1.55	1.62	1.73
	(5%,10%]	1.37	1.38	1.31	1.22	1.03	0.82	0.99	0.98	1.35	1.48	1.52	1.69
	(10%,15%]	1.47	1.38	1.38	1.29	1.15	0.93	1.08	1.18	0.97	1.46	1.55	1.68
	(15%,20%]	1.57	1.52	1.49	1.27	1.18	1.01	1.13	1.19	1.27	0.86	1.45	1.61
	(20%,25%]	1.56	1.46	1.44	1.40	1.22	1.13	1.13	1.20	1.26	1.34	0.76	1.59
	(25%,+∞)	1.71	1.68	1.62	1.45	1.44	1.21	1.24	1.24	1.27	1.28	1.29	1.29

Panel B. China		Focal stock return											
		(-∞,-25%]	(-25%,-20%]	(-20%,-15%]	(-15%,-10%]	(-10%,-5%]	(-5%,0]	(0,5%]	(5%,10%]	(10%,15%]	(15%,20%]	(20%,25%]	(25%,+∞)
Portfolio return	(-∞,-25%]	0.00	0.30	0.38	0.47	0.67	1.04	1.53	1.58	1.51	1.37	1.31	1.02
	(-25%,-20%]	0.23	0.30	0.50	0.61	0.78	1.10	1.53	1.65	1.59	1.43	1.42	1.19
	(-20%,-15%]	0.25	0.37	0.45	0.63	0.82	1.11	1.54	1.64	1.58	1.53	1.31	1.10
	(-15%,-10%]	0.29	0.37	0.48	0.58	0.84	1.13	1.54	1.62	1.54	1.44	1.37	1.16
	(-10%,-5%]	0.32	0.48	0.49	0.65	0.86	1.16	1.54	1.62	1.52	1.50	1.46	1.07
	(-5%,0]	0.55	0.64	0.74	0.78	1.03	1.27	1.61	1.66	1.60	1.43	1.41	1.04
	(0,5%]	0.86	0.92	0.90	0.98	1.17	1.41	1.61	1.64	1.57	1.45	1.31	1.03
	(5%,10%]	1.00	1.02	1.06	1.09	1.28	1.39	1.69	1.57	1.56	1.48	1.36	1.14
	(10%,15%]	1.14	1.08	1.14	1.19	1.42	1.38	1.68	1.69	1.44	1.46	1.41	1.20
	(15%,20%]	1.24	1.04	1.21	1.20	1.36	1.39	1.69	1.68	1.62	1.29	1.41	1.24
	(20%,25%]	1.03	1.16	1.10	1.15	1.44	1.39	1.68	1.70	1.63	1.54	1.19	1.23
	(25%,+∞)	0.95	1.06	1.15	1.14	1.37	1.39	1.69	1.66	1.57	1.50	1.43	1.14

Figure 3. The disposition effect and past market returns

This figure presents the time series of past market returns and the disposition effect across all investors. We estimate the quarterly disposition effect by estimating the coefficients of the interaction terms between the *Gain* dummy and quarterly dummies in a Cox hazard regression model. The first panel shows the US sample results, where the past market return is measured over the past eight quarters. The second panel shows the Chinese sample results, where the past market return is measured over the past three quarters. The left-side of the y-axis is the disposition effect. The right-side of the y-axis is the past market return. The x-axis is quarter.

Panel A. US



Panel B. China

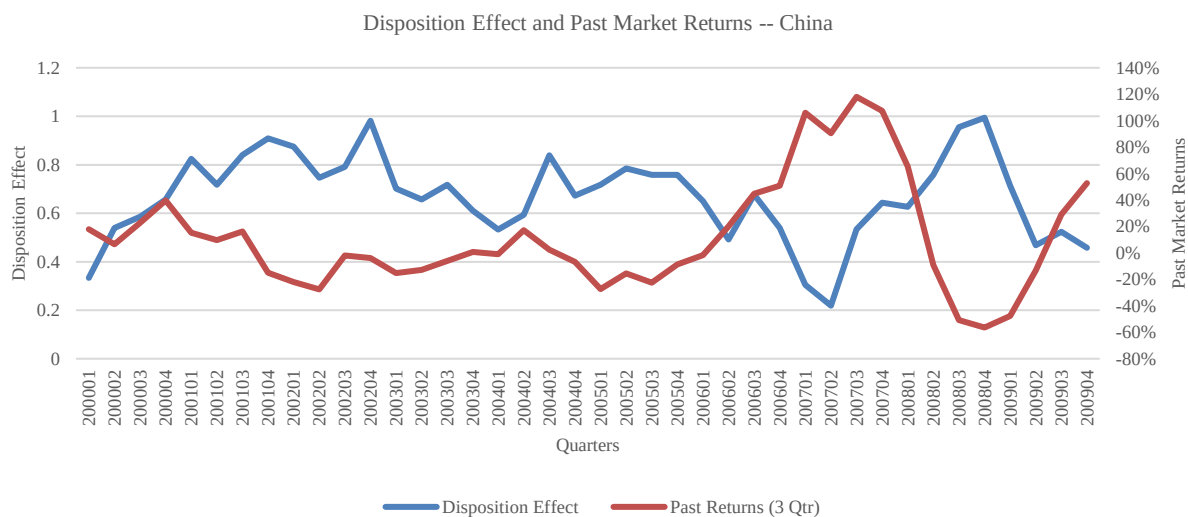


Table 1. Summary statistics

This table presents the summary statistics for the retail brokerage samples: the US sample (Panel A) and the Chinese sample (Panel B). We group all observations into four categories by the values of *Gain* and *Portfolio Gain*. For each group, we report the mean and median for a few portfolio and stock characteristics. *Gain* is a dummy variable that equals 1 if the current price of a stock is higher than its purchase price after adjusting for splits and dividends, and 0 otherwise. *Ret* is the holding period return of an investor-stock-day. *Portfolio Gain* is a dummy variable that equals 1 if the portfolio is at a gain, and 0 otherwise. *Portfolio return* is calculated as the total dollar gains/losses across all stocks held by an investor at the end of day t , divided by the total purchase costs of these stocks. *Time owned* is the number of trading days since purchase. *Volatility* is the standard deviation of daily returns, calculated using the 250 days prior to the purchase. The last three rows report the number of observations (in millions), the number of sell observations, and the daily propensity to sell.

Panel A. US

	Mean				Median			
	Yes	No	Yes	No	Yes	No	Yes	No
Gain	Yes	No	Yes	No	Yes	No	Yes	No
Portfolio Gain	Yes	No	Yes	No	Yes	No	Yes	No
Ret	0.488	0.214	-0.192	-0.270	0.264	0.117	-0.126	-0.210
Portfolio ret	0.326	-0.108	0.222	-0.180	0.209	-0.075	0.126	-0.138
Time owned	416	293	335	309	315	200	227	225
Volatility	0.025	0.029	0.031	0.035	0.021	0.024	0.027	0.031
Obs. (in millions)	47.03	11.86	20.56	31.09				
Sell Obs.	112,959	47,713	40,819	58,392				
% Sell	0.240	0.402	0.199	0.188				

Panel B. China

	Mean				Median			
	Yes	No	Yes	No	Yes	No	Yes	No
Gain	Yes	No	Yes	No	Yes	No	Yes	No
Portfolio Gain	Yes	No	Yes	No	Yes	No	Yes	No
Ret	0.194	0.096	-0.083	-0.293	0.081	0.045	-0.040	-0.237
Portfolio ret	0.150	-0.082	0.068	-0.211	0.079	-0.043	0.038	-0.175
Time owned	37	41	43	87	25	28	22	67
Volatility	0.041	0.039	0.044	0.038	0.036	0.035	0.037	0.034
Obs. (in millions)	13.48	8.64	5.37	57.31				
Sell Obs.	1,177,794	565,688	271,024	1,053,502				
% Sell	8.736	6.547	5.051	1.838				

Table 2. Baseline regressions

This table reports the baseline Cox proportional hazard regression estimation for equation (2). Panel A (Panel B) shows results using the US (Chinese) sample. The dependent variable is a dummy variable that equals 1 if there is sale (including partial sale) on day $t+1$, and 0 otherwise. *Gain* is a dummy variable that equals 1 if a stock in an investor's portfolio has a positive return since purchase at day t , and 0 otherwise. *Portfolio Gain* is a dummy variable that equals 1 if an investor's portfolio has a positive paper gain at day t , and 0 otherwise. We stratify the baseline hazard function by date, stock, and account in columns 2, 3, and 4, respectively. In column 5, we control for the V-shaped disposition effect (Ben-David and Hirshleifer, 2012) by including 52 return bracket indicators for the focal stock's return: $(-\infty, -50\%)$, ..., $[-4\%, -2\%)$, $[-2\%, 0)$, $[0, 2\%)$, $[2\%, 4\%)$, ..., $[50\%, \infty)$. Column 6 reports results using an alternative definition of *Portfolio Gain*; in that specification, portfolio return is measured without incorporating the return of the stock associated with the given observation. Standard errors are clustered by account. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A. US

	(1)	(2)	(3)	(4)	(5)	(6) rest of portfolio return
Gain	0.833*** (65.69)	0.835*** (66.95)	0.909*** (69.03)	0.970*** (65.33)		1.018*** (69.02)
Portfolio Gain	0.172*** (11.89)	0.136*** (9.14)	0.194*** (13.99)	0.716*** (50.77)	0.704*** (49.33)	0.453*** (34.87)
Gain*Portfolio Gain	-0.584*** (-37.71)	-0.616*** (-39.79)	-0.583*** (-37.78)	-0.809*** (-45.28)	-0.863*** (-48.30)	-0.761*** (-43.75)
Stratified by date	No	Yes	No	No	No	No
Stratified by stock	No	No	Yes	No	No	No
Stratified by account	No	No	No	Yes	Yes	Yes
Return bracket FEs	No	No	No	No	Yes	No
Pseudo-R ²	0.003	0.007	0.008	0.023	0.026	0.021
Obs.	110,554,055	110,554,055	110,554,055	110,554,055	110,554,055	110,554,055

Panel B. China

	(1)	(2)	(3)	(4)	(5)	(6) rest of portfolio return
Gain	0.626*** (132.45)	0.600*** (131.53)	0.618*** (134.14)	0.748*** (152.01)		0.725*** (159.73)
Portfolio Gain	0.362*** (63.45)	0.269*** (44.26)	0.362*** (67.56)	0.463*** (112.55)	0.188*** (50.21)	0.410*** (111.98)
Gain*Portfolio Gain	-0.259*** (-42.54)	-0.187*** (-29.24)	-0.258*** (-44.87)	-0.441*** (-93.68)	-0.146*** (-32.64)	-0.291*** (-67.80)
Stratified by date	No	Yes	No	No	No	No
Stratified by stock	No	No	Yes	No	No	No
Stratified by account	No	No	No	Yes	Yes	Yes
Return bracket FEs	No	No	No	No	Yes	No
Pseudo-R ²	0.004	0.004	0.007	0.013	0.013	0.014
Obs.	84,793,767	84,793,767	84,793,767	84,793,767	84,793,767	84,793,767

Table 3. Subsample analysis

This table reports the results from the Table 2 column 4 regression using different subsamples based on the investor's age, gender, trading frequency (above versus below median), and focal-stock holding period, respectively. The dependent variable is a dummy variable that equals 1 if there is sale (including partial sale) on day $t+1$, and 0 otherwise. *Gain* is a dummy variable that equals 1 if a stock in an investor's portfolio has a positive return since purchase at day t , and 0 otherwise. *Portfolio Gain* is a dummy variable that equals 1 if an investor's portfolio has a positive paper gain, and 0 otherwise. All standard errors are clustered by account. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A. US

	Age			Gender		Trading Frequency		Holding period	
	1 to 40	41 to 55	>55	Female	Male	Low	High	1 to 20	1-250
Gain	1.140*** (32.84)	1.069*** (33.67)	0.890*** (41.39)	0.934*** (15.63)	1.012*** (49.44)	0.763*** (23.67)	0.988*** (62.60)	0.853*** (31.85)	0.990*** (59.89)
Portfolio Gain	0.652*** (17.98)	0.752*** (23.01)	0.704*** (35.91)	0.676*** (9.37)	0.732*** (36.87)	0.794*** (23.75)	0.703*** (46.24)	0.459*** (17.86)	0.690*** (43.27)
Gain*Portfolio Gain	-0.716*** (-16.16)	-0.849*** (-22.13)	-0.799*** (-31.44)	-0.697*** (-9.64)	-0.829*** (-33.87)	-0.710*** (-18.36)	-0.809*** (-41.64)	-0.501*** (-16.31)	-0.772*** (-38.43)
Stratified by account	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pseudo-R ²	0.038	0.029	0.018	0.022	0.026	0.016	0.023	0.017	0.023
Obs.	12,524,750	23,819,060	59,511,195	6,226,295	55,641,482	60,205,982	50,348,073	6,765,147	54,493,253

Panel B. China

	Age			Gender		Trading Frequency		Holding period	
	1 to 40	41 to 55	>55	Female	Male	Low	High	1 to 20	1-250
Gain	0.739*** (99.81)	0.769*** (113.74)	0.728*** (74.12)	0.812*** (128.57)	0.699*** (97.78)	1.199*** (176.96)	0.588*** (116.03)	0.572*** (119.45)	0.721*** (149.13)
Portfolio Gain	0.488*** (67.53)	0.474*** (77.80)	0.433*** (54.16)	0.482*** (83.06)	0.456*** (76.72)	0.676*** (86.91)	0.396*** (84.94)	0.317*** (78.71)	0.448*** (111.80)
Gain*Portfolio Gain	-0.472*** (-56.79)	-0.446*** (-62.16)	-0.415*** (-47.66)	-0.460*** (-70.80)	-0.433*** (-58.54)	-0.606*** (-71.30)	-0.365*** (-69.63)	-0.274*** (-60.16)	-0.420*** (-91.48)
Stratified by account	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pseudo-R ²	0.014	0.014	0.012	0.016	0.012	0.039	0.008	0.008	0.013
Obs.	18,114,248	35,762,807	30,916,712	40,733,457	35,283,000	54,211,615	30,582,152	21,422,788	62,911,421

Table 4. Matching sample analysis

This table reports the regression results of the matching analysis. The dependent variable is a dummy variable that equals 1 if there is sale (including partial sale) on day $t+1$, and 0 otherwise. *Gain* is a dummy variable that equals 1 if a stock in an investor's portfolio has a positive return since purchase at day t , and 0 otherwise. *Portfolio Gain* is a dummy variable that equals 1 if an investor's portfolio has a positive paper gain, and 0 otherwise. In columns 1 and 2, we focus on the instances in which there are at least two positions built on the same day and of the same stock. In columns 3 and 4, we further require that the purchase price is the same for the US sample (Panel A). In the Chinese sample (Panel B), we round the purchase price to the nearest eighth and require that the rounded purchase prices are the same. In columns 5 and 6, we further require that the positions are acquired in one purchase, i.e., we only consider (investor, stock, purchase date) triples such that the investor liquidates some of her position in the stock before purchasing any more shares of the stock. In columns 1, 3, and 5, we stratify by account, and in columns 2, 4, and 6, we stratify by position built on the same day and of the same stock. Standard errors are clustered by account. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A. US

	(1)	(2)	(3)	(4)	(5)	(6)
Gain	0.999*** (57.14)	0.641*** (31.89)	1.003*** (45.58)	0.615*** (17.82)	0.952*** (37.72)	0.328*** (6.63)
Portfolio Gain	0.704*** (42.18)	0.084*** (5.79)	0.718*** (31.82)	0.115*** (6.40)	0.681*** (25.32)	0.120*** (5.83)
Gain×Portfolio Gain	-0.831*** (-38.90)	-0.588*** (-34.13)	-0.838*** (-30.26)	-0.550*** (-23.30)	-0.797*** (-24.81)	-0.559*** (-20.78)
Stratification	Account	Positions in same stock with same purchase date	Account	Positions in same stock with same purchase date and price	Account	Positions in same stock with same purchase date and price; requiring one purchase
Pseudo-R ²	0.027	0.008	0.032	0.006	0.030	0.171
Obs.	57,680,893	57,680,893	29,807,432	29,807,432	21,457,460	21,457,460

Panel B. China

	(1)	(2)	(3)	(4)	(5)	(6)
Gain	0.845*** (164.36)	0.451*** (10.86)	0.845*** (164.36)	0.250*** (53.32)	0.896*** (182.59)	0.273*** (54.87)
Portfolio Gain	0.553*** (127.39)	0.233*** (29.20)	0.553*** (127.39)	0.266*** (56.11)	0.589*** (129.69)	0.284*** (56.41)
Gain×Portfolio Gain	-0.510*** (-102.02)	-0.296*** (-18.78)	-0.510*** (-102.02)	-0.178*** (-32.69)	-0.537*** (-102.83)	-0.205*** (-35.70)
Stratification	Account	Positions in same stock with same purchase date	Account	Positions in same stock with same purchase date and price	Account	Positions in same stock with same purchase date and price; requiring one purchase
Pseudo-R ²	0.017	0.000	0.017	0.002	0.021	0.002
Obs.	67,679,549	67,679,549	43,802,363	43,802,363	37,317,624	37,317,624

Table 5. The PDDE in two experimental settings

We report the results for the hazard regressions shown in equation (2) on the experimental samples. The samples consist of all subject-round-stock triples such that the subject owns shares of the given stock and at least one other stock entering the given round. The dependent variable is a dummy variable that equals 1 if there is sale (including partial sale) of the fictitious stock in the given round, and 0 otherwise. *Portfolio Gain* is a dummy variable that equals 1 if a subject's portfolio has a positive paper gain, and 0 otherwise. In column 1, the sample consists of the Chinese university students. Columns 2 and 3 report the results for the MTurk sample and the pooled sample consisting of all subjects (Chinese and MTurk), respectively. In all columns, we stratify by subject. In columns 4-6, we repeat these regressions controlling for the V-shaped disposition effect (Ben-David and Hirshleifer, 2012) by including 52 return bracket indicators for the focal stock's return: $(-\infty, -50\%)$, ..., $[-4\%, -2\%)$, $[-2\%, 0)$, $[0, 2\%)$, $[2\%, 4\%)$, ..., $[50\%, \infty)$. Standard errors are clustered by subject. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
Gain	0.699*** (11.13)	0.491*** (10.63)	0.559*** (15.03)			
Portfolio Gain	0.614*** (13.50)	0.447*** (13.94)	0.503*** (19.12)	0.574*** (13.47)	0.476*** (15.38)	0.512*** (20.32)
Gain*Portfolio Gain	-0.420*** (-7.26)	-0.402*** (-9.49)	-0.406*** (-11.89)	-0.362*** (-6.93)	-0.421*** (-10.71)	-0.400*** (-12.74)
Sample	Chinese	Mturk	Pooled	Chinese	MTurk	Pooled
Stratified by subject	Yes	Yes	Yes	Yes	Yes	Yes
Return bracket FEs	No	No	No	Yes	Yes	Yes
Pseudo-R ²	0.024	0.010	0.014	0.013	0.004	0.016
Obs.	52,903	162,810	215,713	52,903	162,810	215,713

Table 6. Alternative mechanisms

We examine whether alternative mechanisms can explain our main finding. The dependent variable is a dummy variable that equals 1 if there is sale (including partial sale) on day $t+1$, and 0 otherwise. *Gain* is a dummy variable that equals 1 if a stock in an investor's portfolio has a positive return since purchase at day t , and 0 otherwise. *Portfolio Gain* is a dummy variable that equals 1 if an investor's portfolio has a positive paper gain, and 0 otherwise. Following Hartzmark (2015), in Panel A column 1, we add indicator variables for the best and worst performing stocks in an investor's portfolio, and in column 2, we include rank indicator variables for the 15 best and 15 worst performing stocks. In columns 3 and 4, we run the test within tax-exempt accounts and taxable accounts, respectively. In column 5, the dependent variable is defined differently: it is equal to 1 for full liquidations and 0 otherwise. For Panel B, we replicate our analysis in the Chinese sample when possible; the tax incentive columns are omitted because there is no capital gain tax in China during our sample period. All the models are estimated using the Cox proportional hazard model, and the baseline hazard is stratified by account. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A. US

	(1)	(2)	(3)	(4)	(5)
	Rank effect?		Tax incentives?		Rebalancing?
	FE	Extreme	Tax-exempt	Taxable	Full Liquidation
Gain	0.956*** (62.36)	0.831*** (58.70)	0.933*** (36.42)	0.981*** (55.43)	1.008*** (65.86)
Portfolio Gain	0.729*** (50.73)	0.747*** (51.01)	0.732*** (27.48)	0.709*** (42.82)	0.776*** (53.43)
Gain*Portfolio Gain	-0.819*** (-45.43)	-0.768*** (-42.64)	-0.700*** (-21.93)	-0.845*** (-39.99)	-0.915*** (-49.84)
Stratified by account	Yes	Yes	Yes	Yes	Yes
Pseudo-R ²	0.023	0.026	0.025	0.022	0.023
Obs.	110,554,055	110,554,055	32,467,052	78,046,422	118,269,397

Panel B. China

	(1)	(2)	(3)
	Rank effect?		Rebalancing?
	FE	Extreme	Full Liquidation
Gain	0.740*** (143.82)	0.774*** (149.67)	0.896*** (185.17)
Portfolio Gain	0.474*** (112.18)	0.469*** (105.91)	0.608*** (132.46)
Gain*Portfolio Gain	-0.449*** (-94.65)	-0.459*** (-97.98)	-0.597*** (-115.57)
Stratified by account	Yes	Yes	Yes
Pseudo-R ²	0.013	0.014	0.019
Obs.	84,793,767	84,793,767	101,123,567

Table 7. Investor heterogeneity in the disposition effect

We report Cox proportional hazard regression results for specifications that have taken into account potential investor heterogeneity in the disposition effect. The dependent variable is a dummy variable that equals 1 if there is sale (including partial sale) on day $t+1$, and 0 otherwise. *Gain* is a dummy variable that equals 1 if a stock in an investor's portfolio has a positive return since purchase at day t , and 0 otherwise. *Portfolio Gain* is a dummy variable that equals 1 if an investor's portfolio has a positive paper gain, and 0 otherwise. In column 2, portfolio performance is defined in the same way as the baseline specification, in which we only consider unrealized gains/losses. Columns 1 and 3 employ an alternative definition of portfolio performance – besides paper gains/losses of currently held securities, we also add back previously realized gains/losses in the past one year. In all specifications, we control for the V-shaped disposition effect (Ben-David and Hirshleifer, 2012) by including 52 return bracket indicators for the focal stock's return: $(-\infty, -50\%)$, ..., $[-4\%, -2\%)$, $[-2\%, 0)$, $[0, 2\%)$, $[2\%, 4\%)$, ..., $[50\%, +\infty)$. We stratify the baseline hazard function by account in column 1 and by account*gain in columns 2 and 3. Standard errors are clustered by account. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A. US

	(1)	(2)	(3)
	paper gains+ realized gains	paper gains	paper gains+ realized gains
Portfolio Gain	0.511*** (36.05)	0.226*** (15.72)	0.437*** (28.55)
Gain*Portfolio Gain	-0.447*** (-25.43)	-0.126*** (-8.06)	-0.360*** (-21.50)
Stratified by account	Yes	No	No
Stratified by account-gain	No	Yes	Yes
Return bracket FEs	Yes	Yes	Yes
Pseudo-R ²	0.022	0.009	0.010
Obs.	110,554,055	110,554,055	110,554,055

Panel B. China

	(3)	(4)	(5)
	paper gains+ realized gains	paper gains	paper gains+ realized gains
Portfolio Gain	0.152*** (21.04)	0.162*** (41.16)	0.160*** (32.46)
Gain*Portfolio Gain	-0.131*** (-27.67)	-0.150*** (-30.49)	-0.134*** (-27.26)
Stratified by account	Yes	No	No
Stratified by account-gain	No	Yes	Yes
Return bracket FEs	Yes	Yes	Yes
Pseudo-R ²	0.012	0.011	0.011
Obs.	84,793,767	84,793,767	84,793,767

Table 8. The impact of portfolio performance from other household accounts

We report regressions on the comparison of the focal account (the account in which the focal stock resides) portfolio and other accounts within the same household. The dependent variable is a dummy variable that equals 1 if there is any sale (including partial sale) from the end of month t to the end of month $t+1$, and 0 otherwise. *Gain* is a dummy variable that equals 1 if a stock in an investor's portfolio has a positive return since purchase at day t , and 0 otherwise. $AccountPortfolioRet^{all-j}$ is a continuous variable that measures the return for the focal account after excluding the focal stock j , and $HouseholdPortfolioRet^{all-Account}$ is the continuous household return after excluding the entire focal account. In column 2, we restrict the sample to only those households with one adult. To decompose household returns, all returns of subsets of the household are calculated by summing the capital gains within that subset and normalizing them by the cost basis of the entire household. Therefore, the sum of all account returns within a household is equal to the overall household return. We run Cox proportional hazard regressions with account stratification. Standard errors are clustered by account. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)
	All	One adult
Gain	0.685*** (37.49)	0.667*** (13.05)
$AccountPortfolioRet^{all-j}$	1.943*** (22.37)	1.822*** (9.66)
$Gain * AccountPortfolioRet^{all-j}$	-2.302*** (-20.74)	-2.088*** (-9.69)
$HouseholdPortfolioRet^{all-Account}$	1.059*** (7.95)	1.049*** (3.76)
$Gain * HouseholdPortfolioRet^{all-Account}$	-1.364*** (-11.79)	-1.156*** (-4.84)
Stratified by account	Yes	Yes
Testing $\beta_3=\beta_5$, p-value	0.00	0.01
Pseudo- R^2	0.0230	0.0234
Obs.	49,923,314	4,010,386

Table 9. The impact of portfolio performance from various asset classes

We report regression results on the various asset classes of the overall portfolio performance. Assets are classified into five categories: US common stocks, foreign stocks, other stock-type securities, mutual funds, and options. Internet Appendix Table A16 presents details for the asset classifications. $PortfolioRet^{Category}$ is a continuous variable that measures the account's portfolio return for securities in a given category. Performance of securities are calculated following Chang, Solomon, and Westerfield (2016). The dependent variable is a dummy variable that equals 1 if there is any sale (including partial sale) from the end of month t to the end of month $t+1$, and 0 otherwise. $Gain$ is a dummy variable that equals 1 if the stock in question has a positive return since purchase at month end t , and 0 otherwise. $PortfolioRet^{US\ common\ stock-j}$ is the portfolio return for all US common stock after excluding the focal stock j . To decompose portfolio returns, all returns of subsets of the portfolio are calculated by summing the capital gains within that subset and normalizing them by the cost basis of the entire portfolio (from all asset classes). Therefore, the sum of all asset-class returns is equal to the overall portfolio return. We run Cox proportional hazard regressions with account stratification. Standard errors are clustered by account. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)
Category =	Foreign stocks	Other stock-type securities	Mutual funds	Options
Gain	0.356*** (19.06)	0.395*** (16.98)	0.247*** (8.28)	0.303*** (6.46)
$PortfolioRet^{US\ common\ stock-j}$	1.364*** (10.45)	1.505*** (7.95)	1.382*** (5.24)	0.209 (0.71)
$Gain * PortfolioRet^{US\ common\ stock-j}$	-1.770*** (-12.66)	-1.888*** (-9.67)	-1.879*** (-6.59)	-0.839* (-1.86)
$PortfolioRet^{Category}$	0.822*** (3.88)	0.325 (0.82)	1.498** (2.07)	0.675 (1.37)
$Gain * PortfolioRet^{Category}$	-1.093*** (-5.13)	-1.155*** (-3.32)	-0.223 (-0.31)	0.203 (-0.60)
Stratified by account	Yes	Yes	Yes	Yes
Testing $\beta_3 = \beta_5$, p-value	0.01	0.01	0.03	0.29
Pseudo- R^2	0.007	0.008	0.004	0.005
Obs.	461,961	307,688	124,336	26,363

Internet Appendix for

“The Portfolio-Driven Disposition Effect”

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Baolian Wang, and Jared Williams[‡]

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A1. Impact of Unobserved Sophistication/Skill

In Section IV.D of the paper, we analyze the impact of heterogeneity of the disposition effect across investors on the PDDE. We ultimately conclude that it likely has some explanatory power, but it does not fully explain the PDDE. In this section, we explore heterogeneity that is driven by unobserved sophistication/skill. Grinblatt, Keloharju, and Linnainmaa (2012) document that high IQ investors are superior stock pickers and exhibit less of a disposition effect. Hence, it is possible that high IQ investors likely have portfolios at a gain, and low IQ investors (who are prone to the disposition effect and are inferior traders) likely have portfolios at a loss. In other words, it is possible that we are simply documenting a consequence of Grinblatt, Keloharju, and Linnainmaa's (2012) finding. We address this possibility in two ways. First, we use proxies for investor sophistication that have been used by prior researchers to see if our results differ across investors with different levels of sophistication. Second, we decompose an investor's portfolio return into that which is likely driven by skill versus luck, and we compare the moderating effect of portfolio performance in each of these categories. If investor IQ drives our results, the PDDE should be concentrated within the portfolio return driven by skill, while the component of the portfolio return driven by luck should exhibit little or no moderating effect.

The US trading data we use have several demographic characteristics available for a subsample of investors. We follow Dhar and Zhu (2006) in defining an investor's level of sophistication using income and occupation groups. Specifically, we classify investors with an annual income lower than \$40,000 into the low-income category, investors with an annual income between \$40,000 and \$100,000 into the medium-income category, and investors with an annual income more than \$100,000 into the high-income category. We classify individuals as working in "professional" occupations if they report working in "professional/technical" or "managerial/administrative" positions. We classify individuals as working in "nonprofessional" occupations if they report working in "white collar/clerical," "blue collar/craftsman," or "service/sales." Dhar and Zhu (2006) document that investor sophistication is negatively

correlated with the disposition effect, so we test whether the PDDE holds for all subsamples of investors, or if it disappears when we separate investors based on their level of sophistication.

In Internet Appendix Table A12, Panel A, columns 1-5, we report our results on five subsamples based on sophistication-related proxies in the US sample: nonprofessional, professional, low-income, medium-income, and high-income. In columns 1 and 2, we find that the PDDE, as denoted by the interaction coefficient, is nearly identical among professional (-0.800 , t-stat -19.92) and nonprofessional (-0.865 , t-stat -17.46) investors. Therefore, this measure of sophistication does not seem to have any impact on the PDDE. Additionally, in columns 3, 4, and 5, we observe that the PDDE is nearly identical among low-income (-0.852 , t-stat -15.02), medium-income (-0.796 , t-stat -24.87), and high-income (-0.832 , t-stat -20.27) investors. Unfortunately, we do not have employment and income information for the investors in the Chinese sample. Nonetheless, from these results, we conclude that the patterns documented by Dhar and Zhu (2006) have no material impact on the PDDE.

Our final approach is to decompose an investor's portfolio return based on their Daniel, Grinblatt, Titman, and Wermers (DGTW) performance.¹ More specifically, we decompose each investor's portfolio return into two components: one that is the return of the stock's DGTW-matched portfolio, and the other that is the difference between the stock's return and its DGTW-matched portfolio's return, where the DGTW portfolio matching is based on the stock's size, book-to-market, and momentum. The idea is that while highly skilled investors might be able to pick stocks that perform well relative to the stock's matched portfolio, it is unlikely that individual investors can predict the future performance of the market, HML, SMB, and MOM factors. By comparing the effect of portfolio performance that is driven by luck versus skill, we can examine the likelihood that *Portfolio Gain* is simply proxying for investor skill.

¹ See Daniel, Grinblatt, Titman, and Wermers (1997) and Wermers (2004).

We match each stock-date to one of the 125 (5 x 5 x 5) DGTW member groups for each year using the benchmarks available on Russ Wermers' website.² Since the DGTW member groups are created on June 30 of each year, we match all account-stock observations in July-December to the same year and all account-stock observations in January-June to the previous year's member group. With some abuse of terminology, we separate each account-stock-date's return into "alpha" and "beta," where beta represents the return of the corresponding DGTW portfolio (rather than a factor loading on the investor's portfolio) and alpha equals the stock's return minus the matched portfolio. It trivially follows that any stock's cumulative return since the investor purchased it is simply the sum of its alpha and beta.

To separate portfolio return based on alpha and beta performance, we sum all the capital gains within both categories and divide them by the cost basis of the investor's overall portfolio. This procedure generates portfolio-level *Alpha* and *Beta* return variables for each account-date and ensures the sum of *Alpha* and *Beta* is equal to the overall portfolio return. Intuitively, it follows that *Alpha* (*Beta*) is the return for the investor's portfolio that is driven by DGTW alpha (beta) or skill (luck).

Column 6 of Internet Appendix Table A12 tests whether *Portfolio Gain* is simply a proxy for skill using DGTW performance benchmarks. By interacting *Gain* with *Alpha* and *Beta* independently, we can determine if the negative coefficient of *Gain*Portfolio Gain* is driven primarily by one of the two categories that proxy for skill or luck. We observe that the interaction with *Alpha* is comparable in magnitude (−1.547, t-stat −23.9) compared to the interaction of *Gain* and *Beta* (−1.450 t-stat −21.64).³ Similar patterns arise in column 1 of Panel B where we repeat the analysis for the Chinese sample. Thus, the moderating effect of portfolio performance on the disposition effect is present in both *Alpha* and *Beta* categories across both samples. If *Portfolio*

² The DGTW benchmarks are available via <http://terpconnect.umd.edu/~wermers/ftp/Dgtw/coverpage.htm>.

³ It is worth noting that the economic magnitudes of the coefficients are not directly comparable to the other columns, because in this column we use continuous return variables (*Alpha* and *Beta*), while *Portfolio Gain* is a dummy variable in other columns.

Gain were simply a proxy for investor skill, the interaction of *Gain* and *Beta* should have been non-negative. Therefore, we conclude that the PDDE is not simply a manifestation of Grinblatt, Keloharju, and Linnainmaa's (2012) results.

A2. Experiment Details

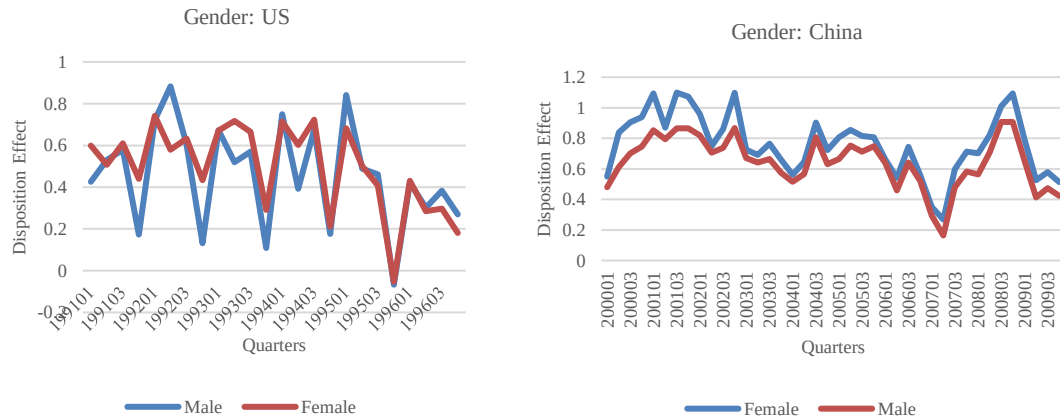
To examine whether the PDDE is robust to different ways that portfolio performance information might be presented to investors, we randomly assigned traders to different experimental treatments, where the portfolio performance information was presented to investors in different manners across the treatments. In some of the treatments, we displayed the sum of all paper gains/losses across all the stocks in the investor's portfolio, while in others, we did not compute an aggregate portfolio performance. In some treatments, we displayed a pop-up window between each trading round that made the portfolio performance more salient to the subject. We describe the display of information in Internet Appendix Table A8 and include an example of what the screen looks like in Internet Appendix Figure A2.

In Internet Appendix Table A9, we examine cross-sectional differences in the PDDE across experimental treatments. We find no significant differences in the PDDE across experimental conditions in the Chinese sample, and very few significant differences across treatments in the MTurk sample, which suggests that the PDDE is robust to different ways of displaying portfolio performance. The lack of difference across conditions may be due to the fact that the PDDE stems from individuals' endogenous framing choice as portfolio performance varies and is thus insensitive to the information environment we present.

Figure A1. Comovement of the disposition effect

This figure presents the comovement of the disposition effect across different groups of investors: by gender (Panel A), by age (Panel B), by portfolio size (Panel C), and by random grouping (Panel D). The y-axis is the average disposition effect within each group of investors and quarter. The x-axis is quarter.

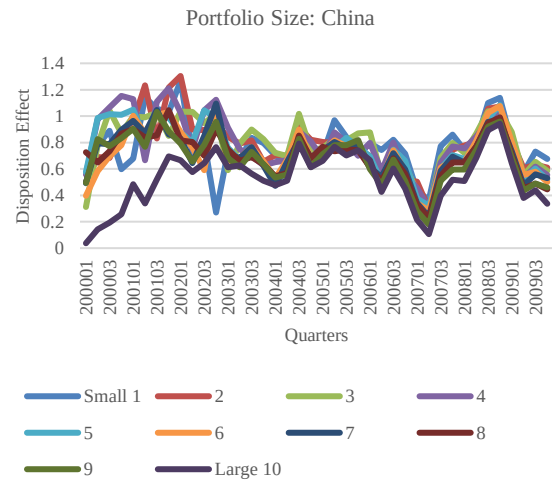
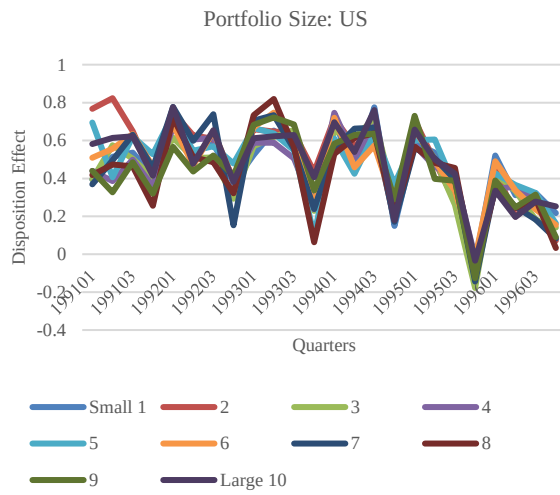
Panel A. by gender



Panel B. by age



Panel C. by portfolio size



Panel D. random

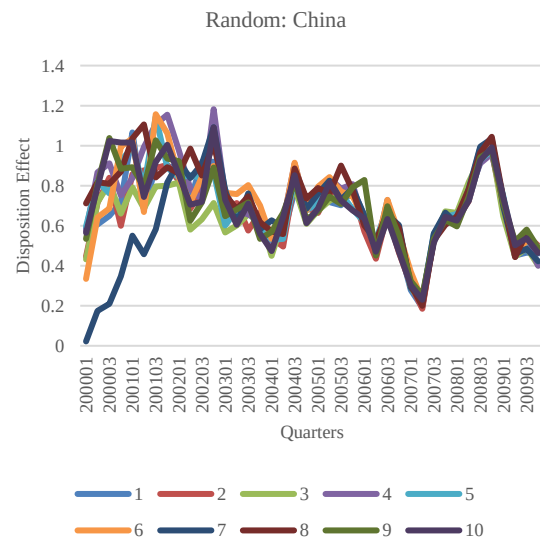
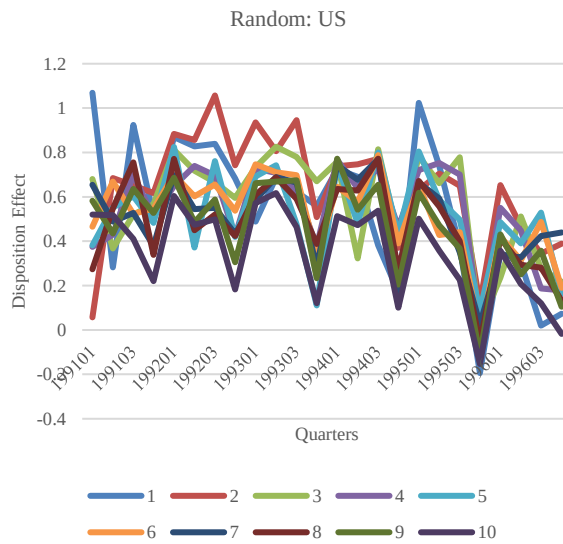


Figure A2. Trading game interface

We present a screenshot of the trading game interface. In this condition, we report the total unrealized gains/losses in the lower right corner.

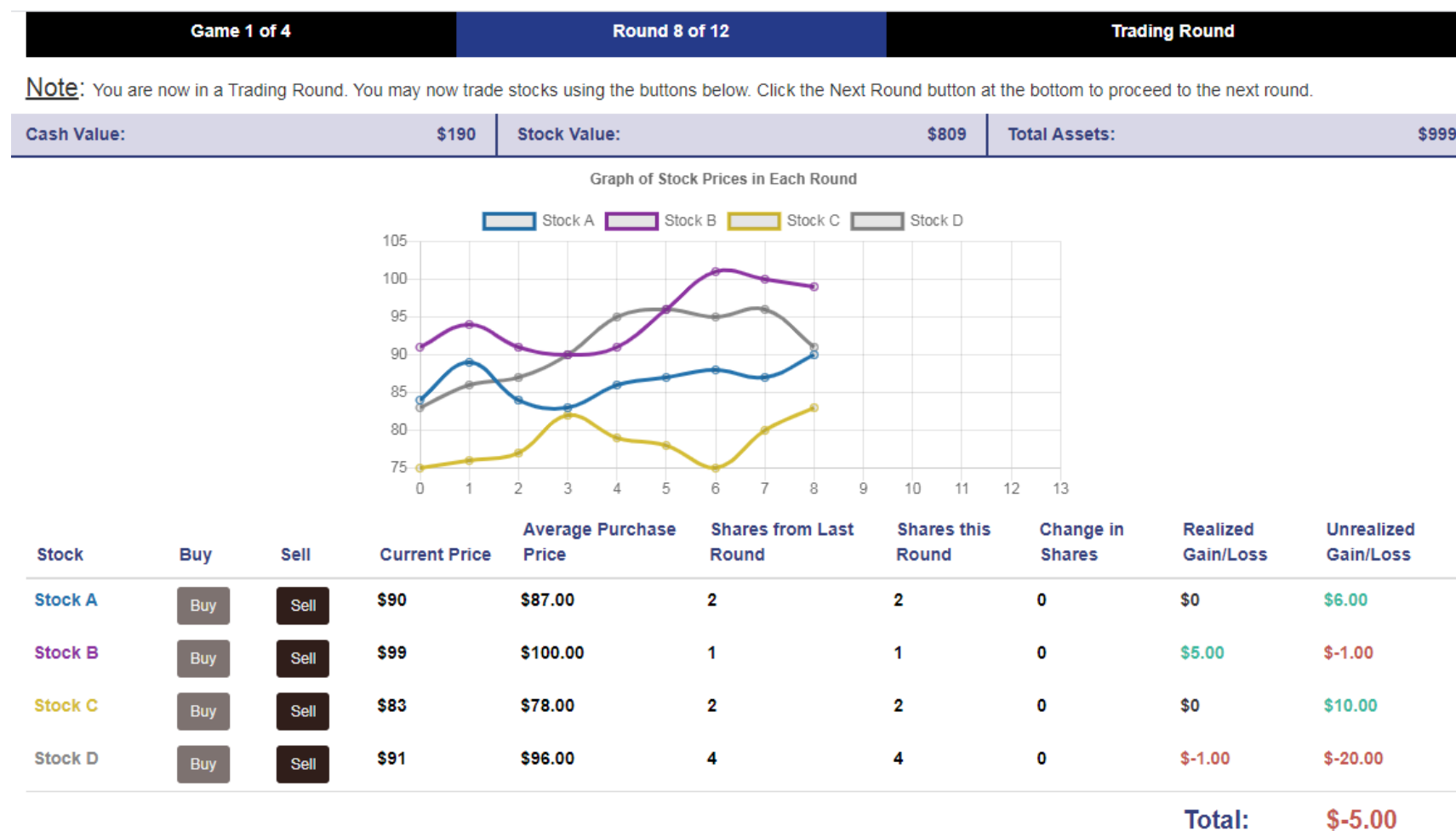


Table A1. Linear probability model, full unconditional sample

This table reports the results for the baseline regressions, similar to Table 2, but using a linear probability model with the full unconditional sample instead of the main methodology, which uses the hazard model described in Section II.B. The dependent variable is a dummy variable that equals 1 if there is sale (including partial sale) on day $t+1$, and 0 otherwise. *Gain* is a dummy variable that equals 1 if a stock in an investor's portfolio has a positive return since purchase at day t , and 0 otherwise. *Portfolio Gain* is a dummy variable that equals 1 if an investor's portfolio has a positive paper gain, and 0 otherwise. All coefficients are multiplied by 100. Standard errors are clustered by account, day, and stock, following the procedure of Cameron, Gelbach, and Miller (2011). ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A. US

	(1)	(2)	(3)	(4)	(5)
Gain	0.216*** (17.49)	0.277*** (22.32)	0.214*** (17.44)	0.249*** (22.30)	0.293*** (24.15)
Portfolio Gain	0.007 (1.55)	0.184*** (20.86)	0.000 (0.07)	0.018*** (4.22)	0.177*** (21.99)
Gain*Portfolio Gain	-0.170*** (-18.16)	-0.202*** (-19.85)	-0.173*** (-18.82)	-0.171*** (-18.79)	-0.206*** (-20.66)
Account FE	No	Yes	No	No	Yes
Date FE	No	No	Yes	No	Yes
Stock FE	No	No	No	Yes	Yes
Adj-R ²	0.000	0.011	0.001	0.001	0.012
Obs.	118,155,489	118,155,489	118,155,489	118,155,489	118,155,489

Panel B. China

	(1)	(2)	(3)	(4)	(5)
Gain	4.153*** (60.23)	3.910*** (62.24)	3.731*** (53.65)	4.072*** (60.96)	3.489*** (59.74)
Portfolio Gain	2.674*** (38.07)	2.303*** (41.19)	1.819*** (26.53)	2.601*** (39.25)	1.462*** (34.59)
Gain*Portfolio Gain	-0.819*** (-10.02)	-0.656*** (-11.18)	-0.720*** (-10.49)	-0.732*** (-9.76)	-0.564*** (-9.90)
Account FE	No	Yes	No	No	Yes
Date FE	No	No	Yes	No	Yes
Stock FE	No	No	No	Yes	Yes
Adj-R ²	0.018	0.053	0.024	0.020	0.059
Obs.	101,123,567	101,123,567	101,123,567	101,123,567	101,123,567

Table A2. Linear probability model, sale-conditioned sample (Odean [1998] specification)

This table reports the results using the methodology from Odean (1998), which restricts to trading days that investors actually sell some stock from their portfolio. The dependent variable is a dummy variable that equals 1 if there is sale (including partial sale) on day $t+1$, and 0 otherwise. *Gain* is a dummy variable that equals 1 if a stock in an investor's portfolio has a positive return since purchase at day t , and 0 otherwise. *Portfolio Gain* is a dummy variable that equals 1 if an investor's portfolio has a positive paper gain, and 0 otherwise. All coefficients are multiplied by 100. Standard errors are clustered by account, day, and stock, following the procedure of Cameron, Gelbach, and Miller (2011). ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A. US

	(1)	(2)	(3)	(4)	(5)
Gain	11.534*** (21.90)	13.619*** (27.38)	11.689*** (22.17)	12.552*** (26.25)	14.169*** (31.34)
Portfolio Gain	-2.286*** (-3.34)	1.901*** (6.85)	-1.851*** (-3.00)	-1.579*** (-3.40)	2.350*** (9.25)
Gain*Portfolio Gain	-4.771*** (-9.76)	-8.482*** (-18.32)	-4.969*** (-10.16)	-5.115*** (-11.41)	-8.670*** (-21.08)
Account FE	No	Yes	No	No	Yes
Date FE	No	No	Yes	No	Yes
Stock FE	No	No	No	Yes	Yes
Adj-R ²	0.012	0.117	0.018	0.037	0.137
Obs.	1,479,983	1,479,983	1,479,983	1,479,983	1,479,983

Panel B. China

	(1)	(2)	(3)	(4)	(5)
Gain	15.012*** (55.05)	16.866*** (78.03)	16.037*** (57.44)	14.727*** (57.60)	17.361*** (85.58)
Portfolio Gain	-1.231*** (-5.32)	-1.904*** (-11.40)	-0.058 (-0.24)	-1.232*** (-5.48)	-0.726*** (-5.49)
Gain*Portfolio Gain	-3.216*** (-10.74)	-6.439*** (-16.59)	-3.990*** (-13.43)	-3.084*** (-10.29)	-6.944*** (-18.95)
Account FE	No	Yes	No	No	Yes
Date FE	No	No	Yes	No	Yes
Stock FE	No	No	No	Yes	Yes
Adj-R ²	0.023	0.097	0.028	0.028	0.103
Obs.	8,184,697	8,184,697	8,184,697	8,184,697	8,184,697

Table A3. Baseline regressions when the portfolio return is measured excluding the stock associated with the observation

This table is the same as columns 1-4 of Table 2, except that portfolio return is measured without considering the return of the stock associated with the given observation. For example, if investor i owns three stocks (A, B, and C) on date t , the portfolio gain variable only considers the performance of stocks B and C for the observation (i, A, t) , and it only considers the performance of stocks A and C for the observation (i, B, t) , etc. The dependent variable is a dummy variable that equals 1 if there is sale (including partial sale) on day $t+1$, and 0 otherwise. *Gain* is a dummy variable that equals 1 if the stock has a positive return at day t since the day that the investor purchased the stock, and 0 otherwise. *Portfolio Gain* is a dummy variable that equals 1 if an investor's portfolio has a positive return, and 0 otherwise. All the models are estimated using Cox proportional hazard model. Standard errors are clustered by account. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A. US

	(1)	(2)	(3)	(4)
Gain	0.804*** (72.48)	1.018*** (69.02)	0.773*** (69.98)	0.882*** (74.09)
Portfolio Gain	0.000 (0.04)	0.453*** (34.87)	-0.055*** (-3.92)	0.030** (2.30)
Gain*Portfolio Gain	-0.557*** (-42.65)	-0.761*** (-43.75)	-0.568*** (-43.38)	-0.558*** (-42.27)
Stratified by account	No	Yes	No	No
Stratified by date	No	No	Yes	No
Stratified by stock	No	No	No	Yes
Pseudo-R ²	0.004	0.021	0.009	0.010
Obs.	110,554,055	110,554,055	110,554,055	110,444,711

Panel B. China

	(1)	(2)	(3)	(4)
Gain	0.675*** (146.36)	0.725*** (159.73)	0.650*** (143.89)	0.664*** (147.98)
Portfolio Gain	0.368*** (74.26)	0.410*** (111.98)	0.278*** (55.52)	0.364*** (77.25)
Gain*Portfolio Gain	-0.276*** (-54.62)	-0.291*** (-67.80)	-0.219*** (-45.56)	-0.265*** (-54.84)
Stratified by account	No	Yes	No	No
Stratified by date	No	No	Yes	No
Stratified by stock	No	No	No	Yes
Pseudo-R ²	0.004	0.014	0.004	0.007
Obs.	84,793,767	84,793,767	84,793,767	84,793,767

Table A4. Cox hazard model, alternative definitions of portfolio gain

This table is the same as column 2 of Table 2, except with alternate definitions of portfolio performance for the US (columns 1 and 2) and Chinese (columns 3 and 4) brokerage samples. The dependent variable is a dummy variable that equals 1 if there is sale (including partial sale) on day $t+1$, and 0 otherwise. *Gain* is a dummy variable that equals 1 if a stock in an investor's portfolio has a positive return since purchase at day t , and 0 otherwise. In columns 1 and 3, *Portfolio Gain* is a dummy variable that equals 1 if the number of winners (or gains) is more than the number of losers (or losses), and zero otherwise. In columns 2 and 4, *Portfolio Gain* is a continuous variable measuring the fraction of winners in a portfolio. All the models are estimated using Cox proportional hazard model. Standard errors are clustered by account. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	US		China	
	(1) Portfolio Gain= (#winners>#losers)	(2) Portfolio Gain= #winners/(#losers+#winners)	(3) Portfolio Gain= (#winners>#losers)	(4) Portfolio Gain= #winners/(#losers+#winners)
Gain	0.786*** (59.92)	1.307*** (64.85)	0.641*** (152.67)	0.731*** (131.08)
Portfolio Gain	0.632*** (42.71)	1.460*** (55.97)	0.350*** (72.21)	0.790*** (107.66)
Gain*Portfolio Gain	-0.672*** (-39.86)	-1.754*** (-50.68)	-0.234*** (-46.38)	-0.623*** (-74.79)
Stratified by account	Yes	Yes	Yes	Yes
Pseudo-R ²	0.019	0.023	0.013	0.014
No. of Obs.	110,554,055	110,554,055	84,793,767	84,793,767

Table A5. Cox hazard model, more control variables

This table is the same as column 2 of Table 2, except that we add more control variables. The dependent variable is a dummy variable that equals 1 if there is sale (including partial sale) on day $t+1$, and 0 otherwise. *Gain* is a dummy variable that equals 1 if a stock in an investor's portfolio has a positive return since purchase at day t , and 0 otherwise. *Portfolio Gain* is a dummy variable that equals 1 if an investor's portfolio has a positive return, and 0 otherwise. *Log(Buy price)* is the logged purchase price (in dollars). *Volatility* (*Volatility*⁺) is the stock volatility calculated using daily returns using the 250 days prior to the purchase if the return since purchase is negative (positive), zero otherwise. In the last two columns, we control for return bracket fixed effects. Specifically, we split observations by holding period return into 50 brackets: $(-\infty, -50\%)$, ..., $[-4\%, -2\%)$, $[-2\%, 0)$, $[0, 2\%)$, $[2\%, 4\%)$, ..., $[50\%, \infty)$. Standard errors are clustered by account. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A. US

	(1)	(2)	(3)	(4)	(5)
Gain	0.972*** (65.34)	0.696*** (34.94)	0.698*** (35.05)		
Portfolio Gain	0.716*** (50.80)	0.702*** (49.70)	0.702*** (49.73)	0.704*** (49.33)	0.753*** (51.02)
Gain* Portfolio Gain	-0.809*** (-45.28)	-0.791*** (-44.12)	-0.791*** (-44.12)	-0.863*** (-48.30)	-0.945*** (-50.76)
Log(Buy price)	0.008*** (3.33)		0.013*** (5.55)		0.022*** (9.20)
Volatility ⁻		-2.928*** (-11.19)	-2.814*** (-10.70)		-4.533*** (-16.04)
Volatility ⁺		5.277*** (17.02)	5.423*** (17.28)		3.362*** (11.62)
Stratified by account	Yes	Yes	Yes	Yes	Yes
Return bracket FE	No	No	No	Yes	Yes
Pseudo-R ²	0.023	0.024	0.024	0.026	0.028
Obs.	110,554,013	110,426,251	110,426,209	110,554,055	110,426,209

Panel B. China

	(1)	(2)	(3)	(4)	(5)
Gain	0.746*** (151.69)	0.760*** (151.57)	0.758*** (149.79)		
Portfolio Gain	0.463*** (112.40)	0.463*** (112.53)	0.463*** (112.49)	0.203*** (51.56)	0.200*** (50.88)
Gain* Portfolio Gain	-0.439*** (-93.73)	-0.440*** (-93.89)	-0.439*** (-93.70)	-0.139*** (-31.12)	-0.139*** (-31.28)
Log(Buy price)	-0.007*** (-2.96)		-0.008*** (-3.63)		0.032*** (14.04)
Volatility ⁻		0.190*** (17.76)	0.195*** (18.35)		0.244*** (24.57)
Volatility ⁺		-0.074*** (-4.42)	-0.066*** (-3.92)		-0.065*** (-3.87)
Stratified by account	Yes	Yes	Yes	Yes	Yes
Return bracket FE	No	No	No	Yes	Yes
Pseudo-R ²	0.013	0.013	0.013	0.020	0.020
Obs.	84,793,767	84,793,245	84,793,245	84,793,767	84,793,245

Table A6. Cox hazard model, accounts defined at the household level

This table is the same as columns 1-4 of Table 2, except that we conduct the analysis at the household level instead of the account level. The dependent variable is a dummy variable that equals 1 if there is sale (including partial sale) on day $t+1$, and 0 otherwise. *Gain* is a dummy variable that equals 1 if a stock in an investor's portfolio has a positive return since purchase at day t , and 0 otherwise. *Portfolio Gain* is a dummy variable that equals 1 if an investor's portfolio has a positive return, and 0 otherwise. Standard errors are clustered by household. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A. At the household level

	(1)	(2)	(3)	(4)
Gain	0.826*** (62.34)	0.944*** (61.16)	0.824*** (63.15)	0.896*** (64.50)
Portfolio Gain	0.163*** (10.59)	0.687*** (46.94)	0.117*** (7.40)	0.180*** (12.24)
Gain*Portfolio Gain	-0.571*** (-35.15)	-0.762*** (-41.03)	-0.599*** (-36.94)	-0.566*** (-34.88)
Stratified by household	No	Yes	No	No
Stratified by date	No	No	Yes	No
Stratified by stock	No	No	No	Yes
Pseudo-R ²	0.003	0.020	0.007	0.008
Obs.	106,241,871	106,241,871	106,241,871	106,241,871

Panel B. Only households with one account

	(1)	(2)	(3)	(4)
Gain	0.783*** (33.16)	0.957*** (37.03)	0.789*** (34.83)	0.867*** (36.92)
Portfolio Gain	0.107*** (4.46)	0.707*** (27.89)	0.077*** (3.20)	0.138*** (6.05)
Gain*Portfolio Gain	-0.367*** (-12.28)	-0.709*** (-22.32)	-0.403*** (-13.83)	-0.366*** (-12.58)
Stratified by household	No	Yes	No	No
Stratified by date	No	No	Yes	No
Stratified by stock	No	No	No	Yes
Pseudo-R ²	0.004	0.028	0.009	0.011
Obs.	42,123,856	42,123,856	42,123,856	42,123,856

Table A7. Heatmap examples of a binary DE, PDE, and PDDE

This table is meant to aid in the interpretation of Figure 2. We report in Panels A, B, and C how a binary stock-level disposition effect (DE), a “portfolio-level disposition effect” (“PDE”) whereby investors are more likely to sell a stock when their total portfolio is at a gain (irrespective of the performance of the individual stock), and a portfolio-driven disposition effect (PDDE) would visualize in isolation, respectively. Rows indicate different portfolio return brackets, and columns indicate different focal stock return brackets.

Panel A: Binary DE. No PDE. No PDDE

	(-∞,-25%]	(-25%,-20%]	(-20%,-15%]	(-15%,-10%]	(-10%,-5%]	(-5%,0]	(0,5%]	(5%,10%]	(10%,15%]	(15%,20%]	(20%,25%]	(25%,+∞)
(-∞,-25%]	0.00	0.00	0.00	0.00	0.00	0.00	1.00	1.00	1.00	1.00	1.00	1.00
(-25%,-20%]	0.00	0.00	0.00	0.00	0.00	0.00	1.00	1.00	1.00	1.00	1.00	1.00
(-20%,-15%]	0.00	0.00	0.00	0.00	0.00	0.00	1.00	1.00	1.00	1.00	1.00	1.00
(-15%,-10%]	0.00	0.00	0.00	0.00	0.00	0.00	1.00	1.00	1.00	1.00	1.00	1.00
(-10%,-5%]	0.00	0.00	0.00	0.00	0.00	0.00	1.00	1.00	1.00	1.00	1.00	1.00
(-5%,0]	0.00	0.00	0.00	0.00	0.00	0.00	1.00	1.00	1.00	1.00	1.00	1.00
(0,5%]	0.00	0.00	0.00	0.00	0.00	0.00	1.00	1.00	1.00	1.00	1.00	1.00
(5%,10%]	0.00	0.00	0.00	0.00	0.00	0.00	1.00	1.00	1.00	1.00	1.00	1.00
(10%,15%]	0.00	0.00	0.00	0.00	0.00	0.00	1.00	1.00	1.00	1.00	1.00	1.00
(15%,20%]	0.00	0.00	0.00	0.00	0.00	0.00	1.00	1.00	1.00	1.00	1.00	1.00
(20%,25%]	0.00	0.00	0.00	0.00	0.00	0.00	1.00	1.00	1.00	1.00	1.00	1.00
(25%,+∞)	0.00	0.00	0.00	0.00	0.00	0.00	1.00	1.00	1.00	1.00	1.00	1.00

Panel B: Binary PDE. No DE. No PDDE

	(-∞,-25%]	(-25%,-20%]	(-20%,-15%]	(-15%,-10%]	(-10%,-5%]	(-5%,0]	(0,5%]	(5%,10%]	(10%,15%]	(15%,20%]	(20%,25%]	(25%,+∞)
(-∞,-25%]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
(-25%,-20%]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
(-20%,-15%]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
(-15%,-10%]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
(-10%,-5%]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
(-5%,0]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
(0,5%]	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
(5%,10%]	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
(10%,15%]	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
(15%,20%]	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
(20%,25%]	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
(25%,+∞)	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Panel C: Binary PDDE. No DE. No PDE

	(-∞,-25%]	(-25%,-20%]	(-20%,-15%]	(-15%,-10%]	(-10%,-5%]	(-5%,0]	(0,5%]	(5%,10%]	(10%,15%]	(15%,20%]	(20%,25%]	(25%,+∞)
(-∞,-25%]	0.00	0.00	0.00	0.00	0.00	0.00	1.00	1.00	1.00	1.00	1.00	1.00
(-25%,-20%]	0.00	0.00	0.00	0.00	0.00	0.00	1.00	1.00	1.00	1.00	1.00	1.00
(-20%,-15%]	0.00	0.00	0.00	0.00	0.00	0.00	1.00	1.00	1.00	1.00	1.00	1.00
(-15%,-10%]	0.00	0.00	0.00	0.00	0.00	0.00	1.00	1.00	1.00	1.00	1.00	1.00
(-10%,-5%]	0.00	0.00	0.00	0.00	0.00	0.00	1.00	1.00	1.00	1.00	1.00	1.00
(-5%,0]	0.00	0.00	0.00	0.00	0.00	0.00	1.00	1.00	1.00	1.00	1.00	1.00
(0,5%]	1.00	1.00	1.00	1.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00
(5%,10%]	1.00	1.00	1.00	1.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00
(10%,15%]	1.00	1.00	1.00	1.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00
(15%,20%]	1.00	1.00	1.00	1.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00
(20%,25%]	1.00	1.00	1.00	1.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00
(25%,+∞)	1.00	1.00	1.00	1.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00

Table A8. Experimental design and summary statistics

For each of the seven experimental conditions for the sample of Chinese university students (Panel A) and the 10 conditions from Amazon MTurk workers (Panel B), we describe the experimental condition, list the number of subjects in the treatment, the average pay (in Chinese RMB for the sample of Chinese university students, and in USD for the sample of Amazon MTurkers), and the average number of shares that subjects bought/sold across the four iterations of the trading game.

Panel A. Chinese sample

Condition	Pop-up window	Bar graph	Notes	Subjects	Avg Pay	Shares bought	Shares sold
C1	No pop-up window	No bar graph	The total portfolio gain/loss is not displayed on the trading screen	29	20.87	49.3	28.1
C2	Total unrealized gains/losses is always displayed	Total gains/losses (including realized gains) is displayed	Total portfolio gain/loss was bolded and made more salient	26	21.2	55.4	32
C3	Number of winning and losing stocks is displayed	No bar graph		95	20.76	54.1	28.3
C4	Total unrealized gains/losses is always displayed	No bar graph		169	20.8	52.9	29.4
C5	Total unrealized gains/losses is displayed if the portfolio is at a gain	No bar graph		162	21.05	47.3	25
C6	Total unrealized gains/losses is displayed if the portfolio is at a loss	No bar graph		138	21.14	46.7	26.6
C7	No pop-up window	No bar graph		147	20.82	47.1	25.6
All combined				766	20.93	49.6	27.2

Panel B. MTurk sample

Condition	Pop-up window	Bar graph	Notes	Subjects	Avg Pay	Shares bought	Shares sold
M1	No pop-up window	No bar graph	Total portfolio gain/loss was bolded and made more salient	260	3.26	46.5	19
M2	No pop-up window	No bar graph		230	3.23	47.4	20.1
M3	Total unrealized gains/losses is always displayed	No bar graph		221	3.26	57.7	29.2
M4	Total unrealized gains/losses is always displayed	Total gains/losses (including realized gains) is displayed		240	3.23	50.2	19.7
M5	Gains/losses of each individual stock is displayed	Total gains/losses (including realized gains) is displayed		92	3.24	48.5	20.8
M6	Total gains/losses (including realized gains) is always displayed	Total gains/losses (including realized gains) is displayed		138	3.26	43	16.4
M7	Total unrealized gains/losses is displayed if the portfolio is at a gain	Total gains/losses (including realized gains) is displayed		224	3.24	44.4	17
M8	Total unrealized gains/losses is displayed if the portfolio is at a loss	Total gains/losses (including realized gains) is displayed		230	3.26	46.6	19.8
M9	No pop-up window	Total gains/losses (including realized gains) is displayed		621	3.26	45	19
M10	No pop-up window	Total unrealized gains/losses is displayed		121	3.23	44	18.9
All combined				2377	3.25	47.2	19.9

Table A9. Differences in the PDDE across experimental treatments

We report the baseline regressions from Table 2 in the Chinese university sample (Panel A) and the Amazon MTurk sample (Panel B), except we include dummies for each of the treatments. In addition, we interact the dummies with all the other variables, including V-shaped disposition effect control variables which equal the stock return rounded to the nearest 5%. Standard errors are clustered by subject. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A: Chinese sample

	(1)	(2)
Gain	1.208*** (3.79)	
Portfolio Gain	1.158*** (4.26)	0.992*** (3.56)
Gain*Portfolio Gain	-0.931*** (-3.25)	-0.447 (-1.48)
C2*Gain	-0.732 (-1.44)	
C3*Gain	-0.650* (-1.78)	
C4*Gain	-0.813** (-2.34)	
C5*Gain	-0.491 (-1.41)	
C6*Gain	-0.518 (-1.43)	
C7*Gain	-0.264 (-0.75)	
C2*Portfolio Gain	-0.586* (-1.75)	-0.732** (-2.23)
C3*Portfolio Gain	-0.537* (-1.81)	-0.598** (-1.98)
C4*Portfolio Gain	-0.643** (-2.23)	-0.615** (-2.09)
C5*Portfolio Gain	-0.355 (-1.23)	-0.495* (-1.68)
C6*Portfolio Gain	-0.553* (-1.87)	-0.440 (-1.45)
C7*Portfolio Gain	-0.388 (-1.32)	-0.520* (-1.73)
C2*Gain*Portfolio Gain	0.305 (0.87)	0.505 (1.22)
C3*Gain*Portfolio Gain	0.312 (0.95)	0.211 (0.61)
C4*Gain*Portfolio Gain	0.540* (1.72)	0.386 (1.18)
C5*Gain*Portfolio Gain	0.262 (0.84)	0.293 (0.89)
C6*Gain*Portfolio Gain	0.549* (1.70)	0.136 (0.40)
C7*Gain*Portfolio Gain	0.379 (1.20)	0.344 (1.04)
Stratified by	Subject	Subject-gain
Pseudo-R ²	0.030	0.013
Obs.	52,903	52,903

Panel B: MTurk sample

	(1)	(2)
Gain	0.820*** (4.68)	
Portfolio Gain	0.774*** (6.40)	0.488*** (3.88)
Gain*Portfolio Gain	-0.689*** (-4.74)	-0.200 (-1.41)
M2*Gain	-0.0148 (-0.06)	
M3*Gain	-0.143 (-0.63)	
M4*Gain	-0.430* (-1.84)	
M5*Gain	-0.743** (-2.53)	
M6*Gain	-0.766*** (-2.80)	
M7*Gain	-0.0760 (-0.30)	
M8*Gain	-0.629*** (-2.70)	
M9*Gain	-0.0835 (-0.42)	
M10*Gain	0.227 (0.76)	
M2*Portfolio Gain	0.0490 (0.29)	0.202 (1.14)
M3*Portfolio Gain	-0.188 (-1.27)	-0.158 (-1.06)
M4*Portfolio Gain	-0.288* (-1.83)	-0.227 (-1.38)
M5*Portfolio Gain	-0.537*** (-2.91)	-0.392** (-2.10)
M6*Portfolio Gain	-0.730*** (-3.97)	-0.540*** (-2.92)
M7*Portfolio Gain	-0.195 (-1.14)	-0.203 (-1.14)
M8*Portfolio Gain	-0.420*** (-2.78)	-0.311** (-1.97)
M9*Portfolio Gain	-0.170 (-1.24)	-0.0834 (-0.59)
M10*Portfolio Gain	-0.299 (-1.40)	-0.444** (-2.03)
M2*Gain*Portfolio Gain	-0.303 (-1.50)	-0.481** (-2.35)
M3*Gain*Portfolio Gain	-0.0273 (-0.15)	-0.0215 (-0.12)
M4*Gain*Portfolio Gain	0.160 (0.80)	0.116 (0.58)

M5*Gain*Portfolio Gain	0.302 (1.06)	0.258 (0.92)
M6*Gain*Portfolio Gain	0.575** (2.39)	0.443* (1.91)
M7*Gain*Portfolio Gain	0.125 (0.58)	0.148 (0.69)
M8*Gain*Portfolio Gain	0.242 (1.22)	0.218 (1.11)
M9*Gain*Portfolio Gain	0.212 (1.27)	0.106 (0.64)
M10*Gain*Portfolio Gain	0.531** (2.05)	0.667*** (2.61)
Stratified by	Subject	Subject-gain
Pseudo-R ²	0.021	0.013
Obs.	162,810	162,810

Table A10. The relationship between the disposition effect and past market returns

This table reports the time series correlation and regression results on the relationship between the disposition effect and past market returns. The dependent variable is the disposition effect in quarter t , either in the US data (Panel A) or the Chinese data (Panel B). We estimate the quarterly disposition effect by estimating the coefficients of the interaction terms between the *Gain* dummy and quarterly dummies in a Cox hazard regression model. In the regression, we include the quarterly fixed effects. The independent variables (past market returns) are standardized to have a mean of 0 and a standard deviation of 1. R_{t-1} is the market return in the past one quarter. $R_{t-2,t-1}$ is the cumulative past market return from quarter $t-2$ to quarter $t-1$. Other past market returns are similarly defined. The t-values are Newey-West adjusted with 9 lags. The bold typeface indicates the regressions with the highest R^2 .

Panel A. US

	Correlation coefficient	Regressions results		
		Coefficient	t-value	R^2
R_{t-1}	-0.263	-0.052	-1.16	0.069
$R_{t-2,t-1}$	-0.208	-0.041	-0.77	0.043
$R_{t-3,t-1}$	-0.353	-0.070	-1.49	0.125
$R_{t-4,t-1}$	-0.361	-0.071	-1.77	0.130
$R_{t-5,t-1}$	-0.374	-0.074	-1.39	0.140
$R_{t-6,t-1}$	-0.394	-0.078	-2.17	0.155
$R_{t-7,t-1}$	-0.356	-0.070	-2.16	0.127
$R_{t-8,t-1}$	-0.422	-0.083	-3.23	0.178
$R_{t-9,t-1}$	-0.283	-0.056	-1.10	0.080
$R_{t-10,t-1}$	-0.171	-0.034	-0.68	0.029
$R_{t-11,t-1}$	-0.228	-0.045	-1.24	0.052
$R_{t-12,t-1}$	-0.297	-0.059	-1.56	0.088

Panel B. China

	Correlation coefficient	Regressions results		
		Coefficient	t-value	R^2
R_{t-1}	-0.557	-0.098	-6.68	0.310
$R_{t-2,t-1}$	-0.615	-0.108	-6.65	0.378
$R_{t-3,t-1}$	-0.633	-0.112	-7.69	0.401
$R_{t-4,t-1}$	-0.520	-0.092	-4.48	0.271
$R_{t-5,t-1}$	-0.434	-0.077	-3.31	0.189
$R_{t-6,t-1}$	-0.317	-0.056	-2.33	0.101
$R_{t-7,t-1}$	-0.232	-0.041	-1.71	0.054
$R_{t-8,t-1}$	-0.168	-0.030	-1.35	0.028
$R_{t-9,t-1}$	-0.122	-0.021	-1.07	0.015
$R_{t-10,t-1}$	-0.071	-0.012	-0.67	0.005
$R_{t-11,t-1}$	-0.026	-0.005	-0.27	0.001
$R_{t-12,t-1}$	-0.074	-0.013	-0.83	0.006

Table A11. The PDDE in boom and bust periods

This table reports the results from the Table 2 column 4 regression (i.e., using the Cox proportional hazard model and the baseline hazard is stratified by account) in boom and bust markets. A day is defined as a boom market if the value-weighted market return over the preceding 190 trading days is positive, and a bust market otherwise. The dependent variable is a dummy variable that equals 1 if there is sale (including partial sale) on day $t+1$, and 0 otherwise. *Gain* is a dummy variable that equals 1 if a stock in an investor's portfolio has a positive return since purchase at day t , and 0 otherwise. *Portfolio Gain* is a dummy variable that equals 1 if an investor's portfolio has a positive paper gain, and 0 otherwise. Panel A (Panel B) reports the results for the US (Chinese) brokerage sample. Standard errors are clustered by account. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A. US

	(1)	(2)
	Boom	Bust
Gain	0.978*** (64.48)	0.760*** (15.47)
Portfolio Gain	0.710*** (48.70)	0.745*** (11.52)
Gain*Portfolio Gain	-0.812*** (-44.25)	-0.803*** (-11.77)
Stratified by account	Yes	Yes
Pseudo-R ²	0.023	0.025
Obs.	101,686,373	8,867,682

Panel B. China

	(1)	(2)
	Boom	Bust
Gain	0.739*** (122.30)	0.899*** (134.05)
Portfolio Gain	0.445*** (98.08)	0.548*** (77.86)
Gain*Portfolio Gain	-0.411*** (-79.54)	-0.581*** (-71.06)
Stratified by account	Yes	Yes
Pseudo-R ²	40,137,297	44,656,470
Obs.	0.013	0.020

Table A12. Investor sophistication/skill

We examine whether investor sophistication/skill can explain our main finding. The dependent variable is a dummy variable that equals 1 if there is sale (including partial sale) on day $t+1$, and 0 otherwise. *Gain* is a dummy variable that equals 1 if a stock in an investor's portfolio has a positive return since purchase at day t , and 0 otherwise. *Portfolio Gain* is a dummy variable that equals 1 if an investor's portfolio has a positive paper gain, and 0 otherwise. In Panel A columns 1-5, we separate our sample by income levels and occupations, and we follow the specifications in Dhar and Zhu (2006). Column 6 tests if the PDDE is simply capturing the reduced disposition effect of skilled investors. We remove *Portfolio Gain* and replace the equation with *Alpha* and *Beta*. *Alpha* (*Beta*) is the continuous portfolio return generated by DGTW alpha (beta). For Panel B, we replicate column 6 from Panel A for the Chinese sample. We omit investor sophistication proxies because the Chinese brokerage data do not include employment/income information about account holders. All the models are estimated using the Cox proportional hazard model, and the baseline hazard is stratified by account. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A. US

	(1)	(2)	(3)	(4)	(5)	(6)
	Investor sophistication?					Skilled?
	Professional		Income			DGTW
	No	Yes	Low	Medium	High	Portfolios
Gain	0.995*** (23.16)	1.007*** (29.79)	1.039*** (22.31)	1.013*** (39.58)	0.985*** (26.46)	0.442*** (29.95)
Portfolio Gain	0.803*** (20.42)	0.735*** (20.68)	0.721*** (15.87)	0.709*** (26.44)	0.761*** (22.67)	
Gain*Portfolio Gain	-0.865*** (-17.46)	-0.800*** (-19.92)	-0.852*** (-15.02)	-0.796*** (-24.87)	-0.832*** (-20.27)	
Alpha						1.456*** (23.30)
Beta						1.534*** (24.13)
Gain*Alpha						-1.547*** (-23.95)
Gain*Beta						-1.450*** (-21.64)
Stratified by account	Yes	Yes	Yes	Yes	Yes	Yes
Pseudo-R ²	0.025	0.026	0.027	0.026	0.024	0.018
Obs.	13,639,434	21,331,853	10,861,081	32,984,122	18,514,935	64,828,116

Panel B. China

	(1)
	Skilled?
	DGTW
	Portfolios
Gain	0.305*** (55.23)
Alpha	1.326*** (35.28)
Beta	1.490*** (28.67)
Gain*Alpha	-1.783*** (-82.38)
Gain*Beta	-1.698*** (-44.81)
Stratified by account	Yes
Pseudo-R ²	0.015
Obs.	84,793,767

Table A13. Past realized gains

This table reports robustness checks in the US (Panels A and B) and Chinese (Panels C and D) samples. The dependent variable is a dummy variable that equals 1 if there is sale (including partial sale) on day $t+1$, and 0 otherwise. *Gain* is a dummy variable that equals 1 if a stock in an investor's portfolio has a positive return since purchase at day t , and 0 otherwise. *Portfolio Gain* is a dummy variable that equals 1 if an investor's portfolio has a positive return, and 0 otherwise. In addition to tracking the unrealized portfolio gain at a given time point, we add back capital gains realized in a given past horizon as the overall portfolio return, dating back from 1 month to 2 years. Panels A and C report the regressions with no stratification, and Panels B and D stratify by account. Standard errors are clustered by account. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A. No stratification – US

	(1) 1 month	(2) 3 months	(3) 6 months	(4) 1 year	(5) 2 years
Gain	0.917*** (82.69)	0.906*** (85.11)	0.900*** (85.26)	0.874*** (82.36)	0.842*** (76.98)
Portfolio Gain	0.209*** (15.65)	0.261*** (19.64)	0.305*** (22.72)	0.339*** (24.53)	0.355*** (23.76)
Gain*Portfolio Gain	-0.715*** (-51.90)	-0.705*** (-54.28)	-0.696*** (-54.19)	-0.653*** (-50.10)	-0.596*** (-43.56)
Stratified by account	No	No	No	No	No
Pseudo-R ²	0.004	0.003	0.003	0.003	0.003
Obs.	110,554,055	110,554,055	110,554,055	110,554,055	110,554,055

Panel B. Stratified by account – US

	(1) 1 month	(2) 3 months	(3) 6 months	(4) 1 year	(5) 2 years
Gain	0.993*** (70.03)	0.948*** (65.51)	0.914*** (61.33)	0.865*** (55.80)	0.809*** (48.88)
Portfolio Gain	0.646*** (48.19)	0.604*** (44.10)	0.579*** (42.31)	0.541*** (37.72)	0.502*** (32.70)
Gain*Portfolio Gain	-0.794*** (-47.13)	-0.670*** (-40.40)	-0.577*** (-34.12)	-0.464*** (-25.77)	-0.355*** (-18.12)
Stratified by account	Yes	Yes	Yes	Yes	Yes
Pseudo-R ²	0.022	0.021	0.020	0.019	0.018
Obs.	110,554,055	110,554,055	110,554,055	110,554,055	110,554,055

Panel C. No stratification – China

	(1) 1 month	(2) 3 months	(3) 6 months	(4) 1 year	(5) 2 years
Gain	0.645*** (142.37)	0.672*** (146.56)	0.673*** (146.79)	0.654*** (141.96)	0.640*** (136.62)
Portfolio Gain	0.336*** (58.91)	0.245*** (38.03)	0.188*** (26.85)	0.094*** (11.77)	0.029*** (3.18)
Gain*Portfolio Gain	-0.225*** (-40.56)	-0.197*** (-33.42)	-0.166*** (-26.65)	-0.154*** (-22.96)	-0.139*** (-15.11)
Stratified by account	No	No	No	No	No
Pseudo-R ²	0.004	0.003	0.003	0.003	0.003
Obs.	84,793,767	84,793,767	84,793,767	84,793,767	84,793,767

Panel D. Stratified by account – China

	(1) 1 month	(2) 3 months	(3) 6 months	(4) 1 year	(5) 2 years
Gain	0.701*** (149.76)	0.703*** (150.67)	0.704*** (152.31)	0.687*** (150.95)	0.665*** (143.31)
Portfolio Gain	0.384*** (92.81)	0.314*** (68.12)	0.264*** (51.83)	0.166*** (28.16)	0.129*** (20.15)
Gain*Portfolio Gain	-0.252*** (-50.35)	-0.189*** (-33.45)	-0.159*** (-25.70)	-0.137*** (-18.66)	-0.126*** (-13.29)
Stratified by account	Yes	Yes	Yes	Yes	Yes
Pseudo-R ²	0.013	0.013	0.013	0.012	0.012
Obs.	84,793,767	84,793,767	84,793,767	84,793,767	84,793,767

Table A14. Experimental results with subject-gain stratification

We report the results for the hazard regressions shown in equation (2) on the experimental samples with subject-gain stratification. The samples consists of all subject-round-stock triples such that the subject owns shares of the given stock and at least one other stock entering the given round. In column 1, the sample consists of the Chinese university students. Column 2 and 3 report the results for the MTurk sample and the pooled sample consisting of all subjects (Chinese and MTurk), respectively. The dependent variable is a dummy variable that equals 1 if there is sale (including partial sale) of the fictitious stock in the given round, and 0 otherwise. *Portfolio Gain* is a dummy variable that equals 1 if a subject's portfolio has a positive paper gain, and 0 otherwise. In all columns, we control for the V-shaped disposition effect (Ben-David and Hirshleifer, 2012) by including 52 return bracket indicators for the focal stock's return: $(-\infty, -50\%)$, ..., $[-4\%, -2\%)$, $[-2\%, 0)$, $[0, 2\%)$, $[2\%, 4\%)$, ..., $[50\%, +\infty)$. Standard errors are clustered by subject. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)
Portfolio Gain	0.475*** (10.13)	0.330*** (9.65)	0.376*** (13.51)
Gain*Portfolio Gain	-0.178*** (-2.99)	-0.118*** (-2.67)	-0.132*** (-3.74)
Sample	Chinese	Mturk	Pooled
Stratified by subject-gain	Yes	Yes	Yes
Pseudo-R ²	0.012	0.012	0.011
Obs.	52,903	162,810	215,713

Table A15. The impact of portfolio performance from all household accounts

We report regressions on the moderating effect of the overall household portfolio performance. The dependent variable is a dummy variable that equals 1 if there is any sale (including partial sale) from the end of month t to the end of month $t+1$, and 0 otherwise. *Gain* is a dummy variable that equals 1 if a stock in an investor's portfolio has a positive return since purchase at day t , and 0 otherwise. *HouseholdPortfolioRet^{all}* is a continuous variable that measures the household-level return for all stocks. *HouseholdPortfolioRet^{all-j}* is the continuous household return after excluding the focal stock j . In columns 3 and 4, we restrict the sample to only those households with one adult. We run Cox proportional hazard regressions with account stratification. Standard errors are clustered by account. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)
	All		One adult	
Gain	0.648*** (35.15)	0.684*** (37.14)	0.623*** (12.34)	0.665*** (13.08)
HouseholdPortfolioRet ^{all}	1.571*** (23.42)		1.551*** (9.87)	
Gain* HouseholdPortfolioRet ^{all}	-1.689*** (-24.43)		-1.549*** (-10.17)	
HouseholdPortfolioRet ^{all-j}		1.530*** (22.50)		1.514*** (9.60)
Gain* HouseholdPortfolioRet ^{all-j}		-1.847*** (-25.28)		-1.706*** (-10.69)
Stratified by account	Yes	Yes	Yes	Yes
Pseudo-R ²	0.0224	0.0226	0.0228	0.0230
Obs.	49,923,314	49,923,314	4,010,386	4,010,386

Table A16. Asset class categorization

This table provides detailed information for asset class categorization. *Product Code*, *Security*, and *Product Code 2* are the raw information in the Position Readme file provided by the broker, while *Category* shows our classification based on the similarity of each asset class to US common stocks. *#obs* reports the count of investor-security-month level observations for each asset class and category, and *Category # obs* reports the sum of *#obs* within the given category.

Product	Security	Product	Category	# obs	Category #
COM	US Company Shares	ST	US common stocks	2,882,107,25	2,882,167
FGC	Foreign (Canadian)	ST	Foreign Stocks	3	288,221
FGO	Foreign (Ordinaries)	ST	Foreign Stocks	29,436	
ADR	ADR	ST	Foreign Stocks	151,53	
WAR	Warrants	ST	Other Stock-Type Securities	7,492	234,742
RET	Real Estate Trust	ST	Other Stock-Type	748	
UNI	Units	ST	Other Stock-Type	1,087	
CPR	Convertible Preferred	ST	Other Stock-Type	13,970	
CEM	Closed-End Mutual	ST	Other Stock-Type	158,69	
MLP	Master Limited	ST	Other Stock-Type	33,749	
PRE	Preferred Stock	ST	Other Stock-Type	19,004	
MFS	Mutual Funds (In-House)	MF	Mutual Funds	26,100	319,693
MPL	Marketplace Load	MO	Mutual Funds	1,135	
MPB	Marketplace Load	MO	Mutual Funds	237	
MFB	Bond Mutual Funds	MO	Mutual Funds	10,965	
MFF	One Source Bond	MS	Mutual Funds	29,238	
MFA	One Source Equity	MS	Mutual Funds	203,71	
PFF	Ex One Source Bond	MO	Mutual Funds	1,214	
MFC	Equity Mutual Funds	MO	Mutual Funds	43,192	
PFA	Ex One Source Equity	MO	Mutual Funds	3,893	
OEQ	Option Equity	OP	Options	24,201	28,436
OIN	Options Index	OP	Options	4,235	

Table A17. The impact of portfolio performance from all asset classes

We report regression results on the moderating effect of the overall portfolio performance from all asset classes (including foreign stocks, other stock-type securities, mutual funds, and options). Assets are classified into five categories: US common stocks, foreign stocks, other stock-type securities, mutual funds, and options. Internet Appendix Table A16 presents details for the asset classifications. Performance of securities are calculated following Chang, Solomon, and Westerfield (2016). The dependent variable is a dummy variable that equals 1 if there is any sale (including partial sale) from the end of month t to the end of month $t+1$, and 0 otherwise. *Gain* is a dummy variable that equals 1 if the stock in question has a positive return since purchase at month end t , and 0 otherwise. *PortfolioRet^{all}* is a continuous variable that measures the portfolio return for all securities. *PortfolioRet^{all-j}* is portfolio return after excluding the focal stock j . We run Cox proportional hazard regressions with account stratification. Standard errors are clustered by account. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)
Gain	0.358*** (23.02)	0.369*** (23.82)
PortfolioRet ^{all}	0.390*** (11.87)	
Gain*PortfolioRet ^{all}	-0.368*** (-11.30)	
PortfolioRet ^{all-j}		0.313*** (10.75)
Gain*PortfolioRet ^{all-j}		-0.399*** (-12.81)
Stratified by account	Yes	Yes
Pseudo-R ²	0.006	0.006
Obs.	712,601	712,601

Pre-quiz introductory information

Realized and Unrealized Gain/Loss Description

When trading, there is an important distinction between realized and unrealized gains and losses. In general, when traders buy a stock and its value changes, they have an unrealized gain or loss as long as they have not yet sold that stock. Once they sell the stock, they will have a realized gain or loss.

For example, if Lisa buys 2 shares of Stock A for \$50 each and Stock A goes up in value to \$56, we say Lisa has an unrealized gain of \$12 ($\$6 \text{ per share} \times 2 \text{ shares}$). If Lisa immediately sold her 2 shares of Stock A for \$56 each, she would realize her gain of \$12. At that point we would say Lisa has a realized gain of \$12.

If John bought 3 shares of Stock B for \$80 each and Stock B went down in value to \$71, we say John has an unrealized loss of \$27 ($\$9 \text{ per share} \times 3 \text{ shares}$). If John immediately sold 2 shares of Stock B for \$71 each, he would have a realized loss of \$18 ($\$9 \text{ per share} \times 2 \text{ shares}$). He would also have an unrealized loss of \$9 because he still has not sold 1 share of Stock B which is at a loss of \$9.

Click the Start Quiz button to take an eligibility quiz. You must answer all 3 questions correctly to move on to the trading game (and be eligible for the bonus).

Quiz questions

Question 1

Frank purchased 2 shares of Stock C for \$60 each and 1 share of Stock D for \$80. Stock C went up in value to \$66 and Stock D went down in value to \$73. Frank sold his share of Stock D but did not sell any shares of Stock C. Which of the following is correct?

- A. Frank has a realized gain of \$6 and an unrealized loss of \$7
- B. Frank has a realized gain of \$12 and an unrealized loss of \$7
- C. Frank has an unrealized gain of \$6 and a realized loss of \$7
- D. Frank has an unrealized gain of \$12 and a realized loss of \$7

Question 2

Amy and Bob are two players of the stock trading game. At the end of the game, Amy ends up with cash value of \$700 and stock value of \$540. Bob ends up with cash value of \$1090 and a stock value of \$0. Based on the instructions you read, who will get a higher bonus according to his/her performance?

- A. Amy
- B. Bob

Question 3

Joe starts with \$1000 cash. He first purchases 5 shares of stock A for \$80 each. In a later round, the price of stock A increases to \$90, and he purchases 5 additional shares for a total of 10 shares of Stock A (now with an average purchase price of \$85 each). In the final round, the price of stock A rises again to \$100 per share, and he sells 6 shares. Which of the following is correct at the end of the game?

- A. Joe has \$400 cash (which includes an unrealized gain of \$90) and \$750 of Stock A (which includes a realized gain of \$60)
- B. Joe has \$750 cash (which includes a realized gain of \$90) and \$400 of Stock A (which includes an unrealized gain of \$60)
- C. Joe has \$400 cash (which includes a realized gain of \$60) and \$750 of Stock A (which includes an unrealized gain of \$90)
- D. Joe has \$750 cash (which includes a realized gain of \$60) and \$400 of Stock A (which includes an unrealized gain of \$90)

Experimental treatments

Subjects were randomly assigned to treatments that differed in the manner in which portfolio performance information was presented to the subjects. We conducted 7 treatments for the sample of Chinese university students, and 10 treatments for the Amazon MTurkers. The graphical interface differed across treatments along a few dimensions. First, in some treatments, there was a “pop-up” window that appeared between rounds displaying the investor’s portfolio performance. Within this group of treatments, there was variation in the necessary conditions for the window to pop up: In some conditions, the pop-up window would appear after every round (regardless of the portfolio performance), whereas in other conditions, it would only appear if the portfolio was at a gain, and in others, it would only appear if the portfolio was at a loss. The treatments also differed in whether a bar graph would appear in the trading screen making the subject’s trading performance more salient. In most treatments with a bar graph, the bar graph would display the trader’s total gains/losses (including realized gains/losses), but in one condition (M10), the bar graph displayed the portfolio’s total unrealized gains/losses. Finally, treatments differed in how the trading gains/losses were aggregated in the trading screen. In Internet Appendix Figure A1, notice that the unrealized gains/losses (–\$5) is displayed in the lower right corner of the trading screen. Some treatments did not display this total, and in some treatments, the total was made more salient (by being displayed in bold and large font).

We describe each of the 17 treatments in Internet Appendix Table A8, including the way that information was displayed to the subjects, as well as the number of subjects who participated in the treatment, the average number of shares that subjects bought/sold in the treatment, and the average compensation.